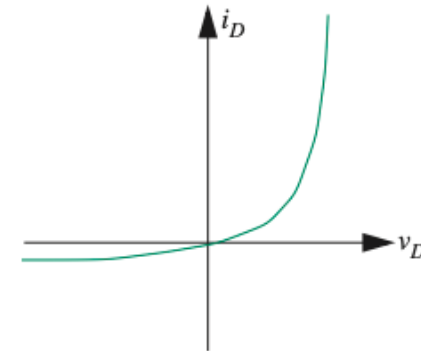
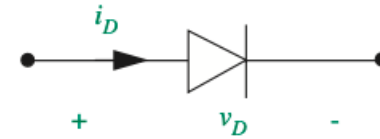
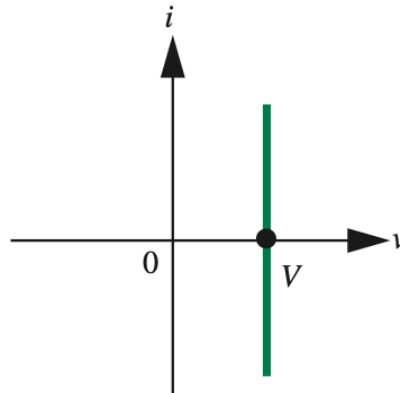
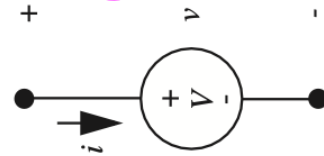
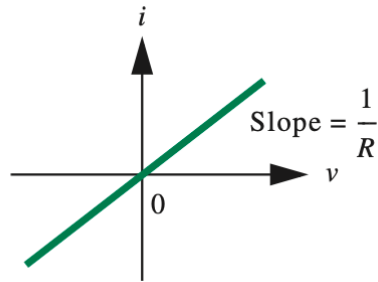
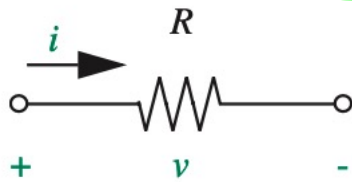


CSE 251 (After Mid)
Electronic Devices and Circuits

Iftexhar Hossain Rahi
22201168

Three terminal devices

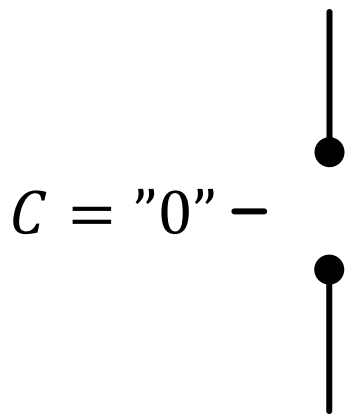
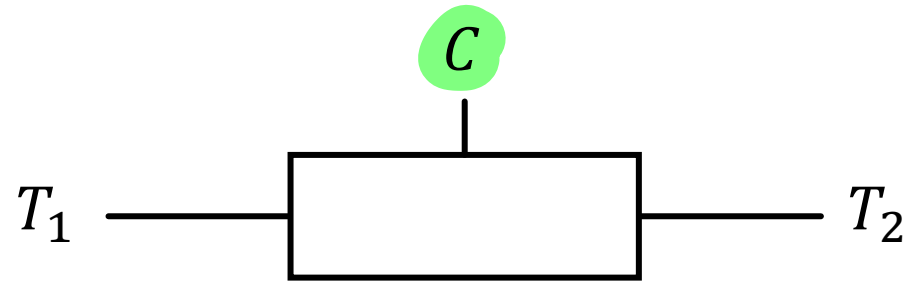
- Two terminal devices have fixed IV characteristics



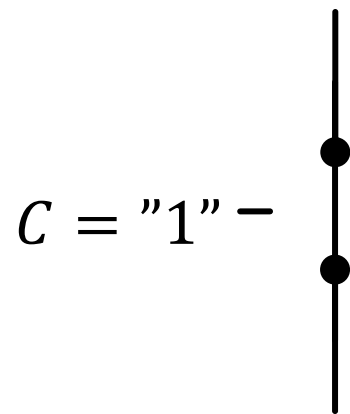
- Three terminal devices – IV of two terminal can be controlled using the third terminal.
- Examples – Switch (linear), Transistors (non-linear)

Switch – IV Characteristics

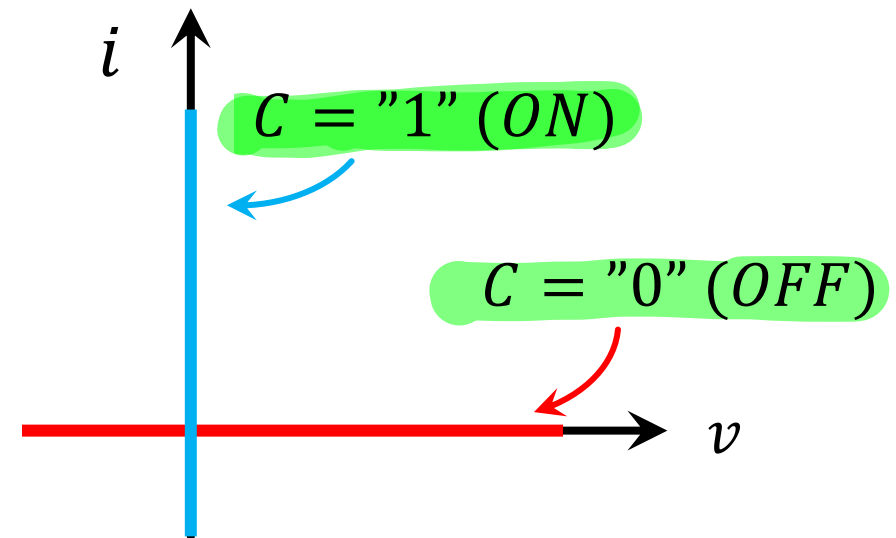
- IV characteristics between terminal T_1 and T_2 is controlled by C



Switch OFF

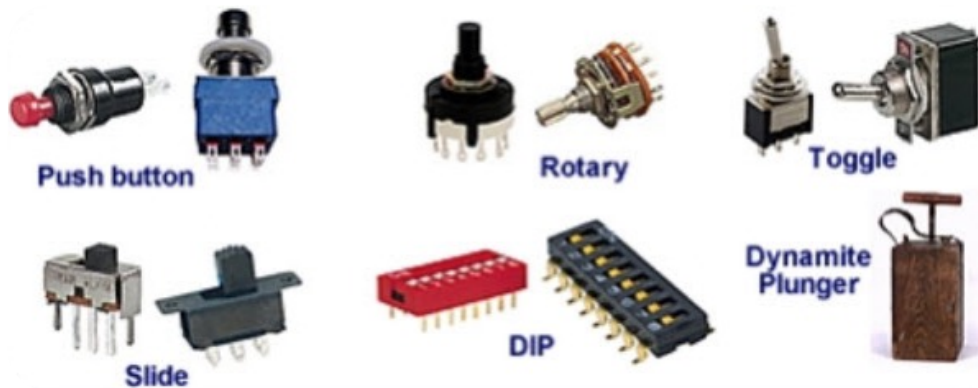


Switch ON

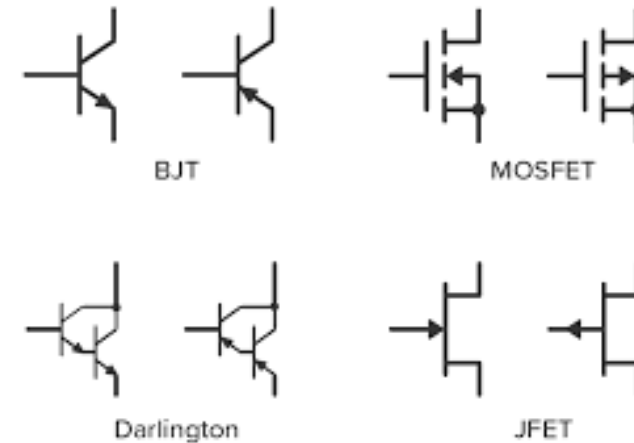


Switch – Types

- Depending on the control, the switch can be
 - **Analog:** Controlled using physical toggle/button
 - **Digital:** Controlled using voltage or current. Example – MOSFET (voltage controlled), BJT (current controlled)



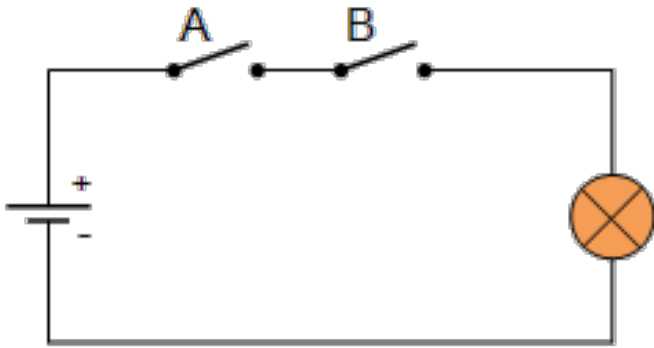
Analog switches



Digital switches (Transistors)

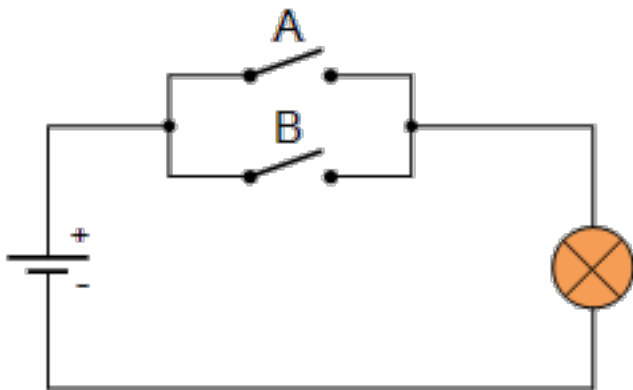
Switch Application – Logic Gates

- We can use switches to build logic gates



A	B	Bulb
0	0	OFF
0	1	OFF
1	0	OFF
1	1	ON

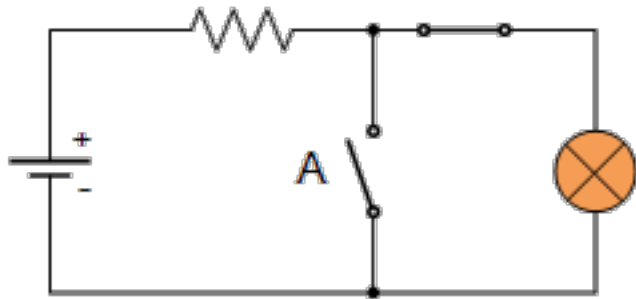
AND operation



A	B	Bulb
0	0	OFF
0	1	ON
1	0	ON
1	1	ON

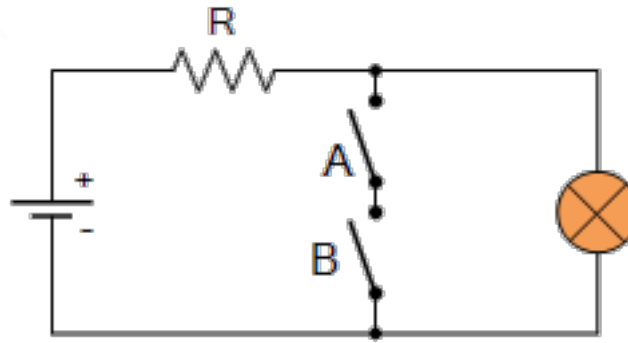
OR operation

Switch Application – Logic Gates



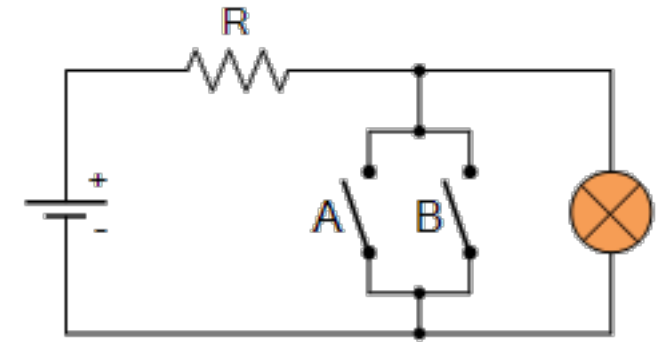
A	Bulb
0	ON
1	OFF

NOT operation



A	B	Bulb
0	0	ON
0	1	ON
1	0	ON
1	1	OFF

NAND operation



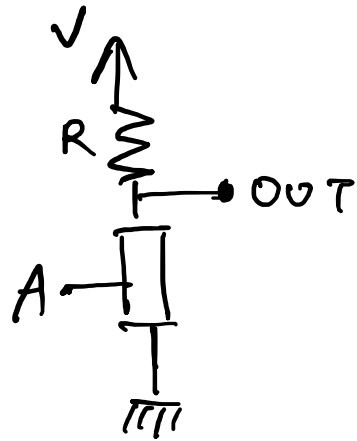
A	B	Bulb
0	0	ON
0	1	OFF
1	0	OFF
1	1	OFF

NOR operation

These circuits are “preferred” – because they can be cascaded to build combinational logic circuits
-> if we remove the bulb and use the voltage across instead to cascade and drive the next gate

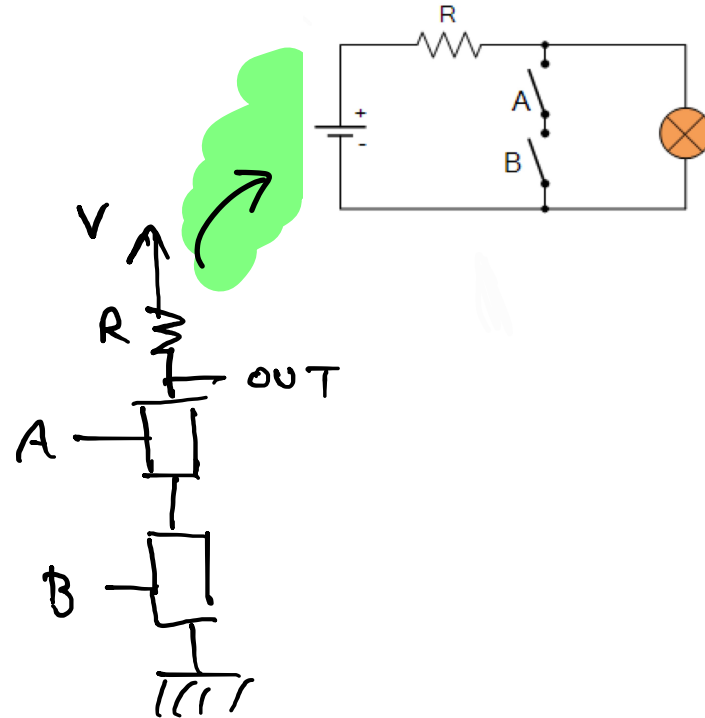
Switch Application – Logic Gates

Alternative representations:



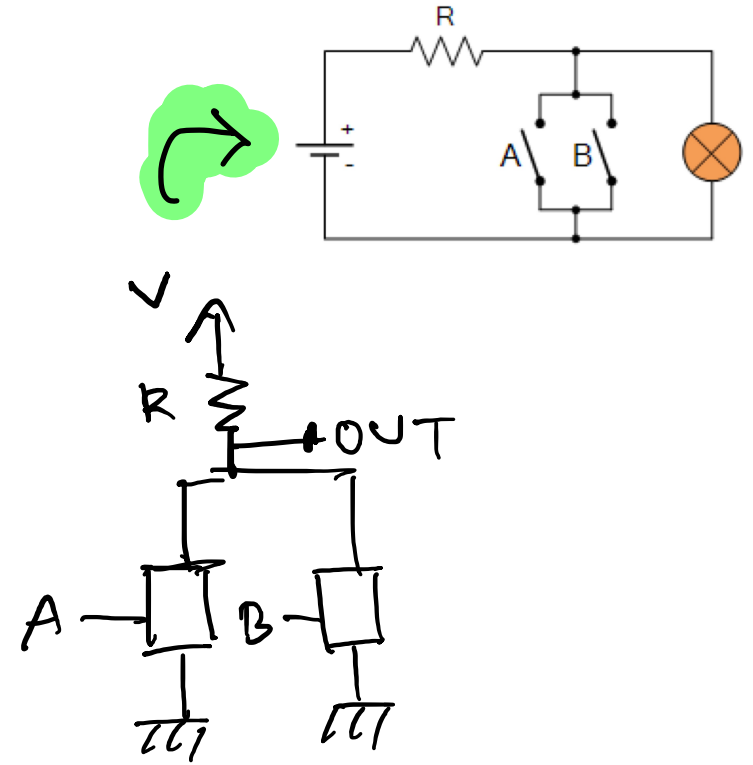
$$\text{OUT} = \overline{A}$$

(NOT)



$$\text{OUT} = \overline{AB}$$

(NAND)

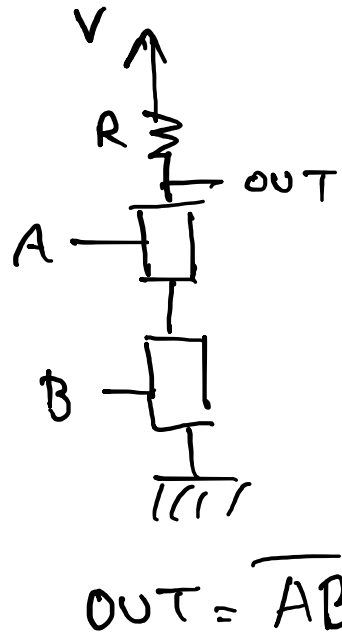


$$\text{OUT} = \overline{A+B}$$

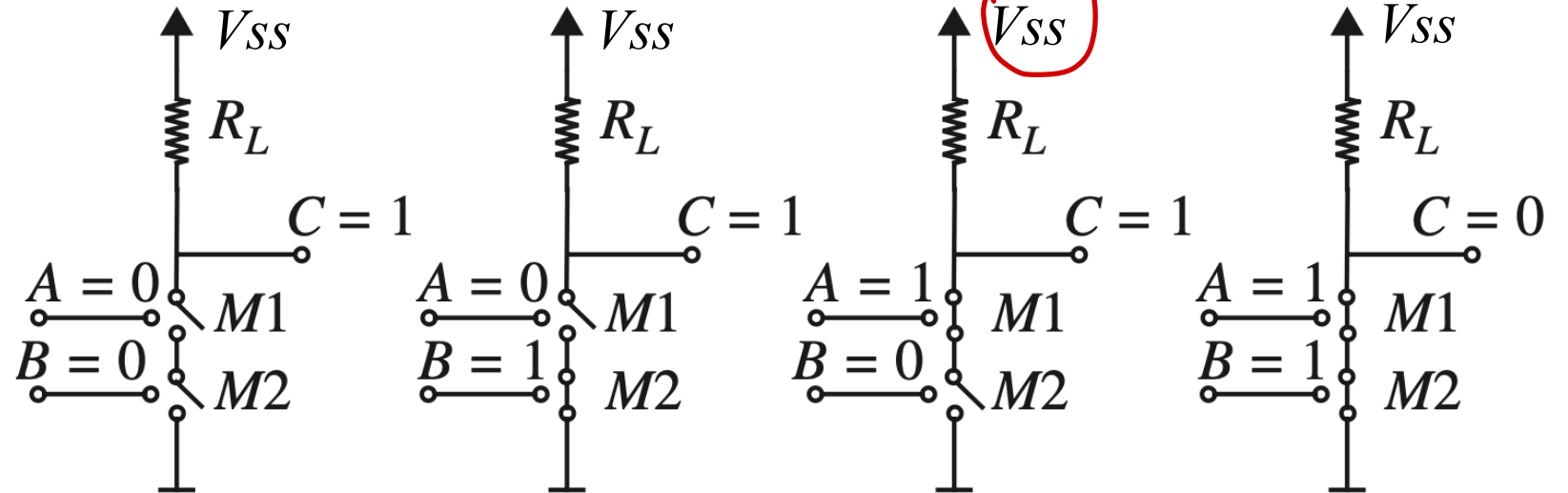
(NOR)

Switch Application – Logic Gates

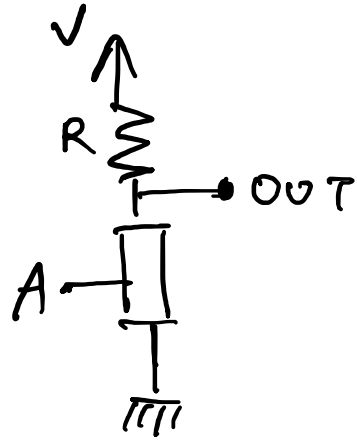
Alternative representations:



(NAND)

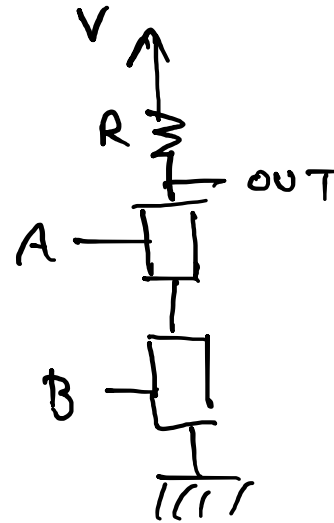


Switch Application – Logic Gates



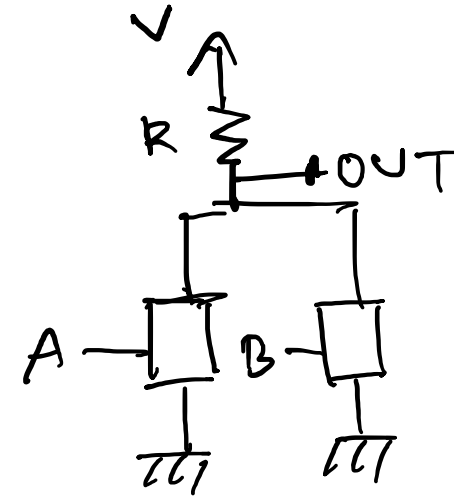
$$OUT = \overline{A}$$

A	V _{OUT}
0	5V
1	0V



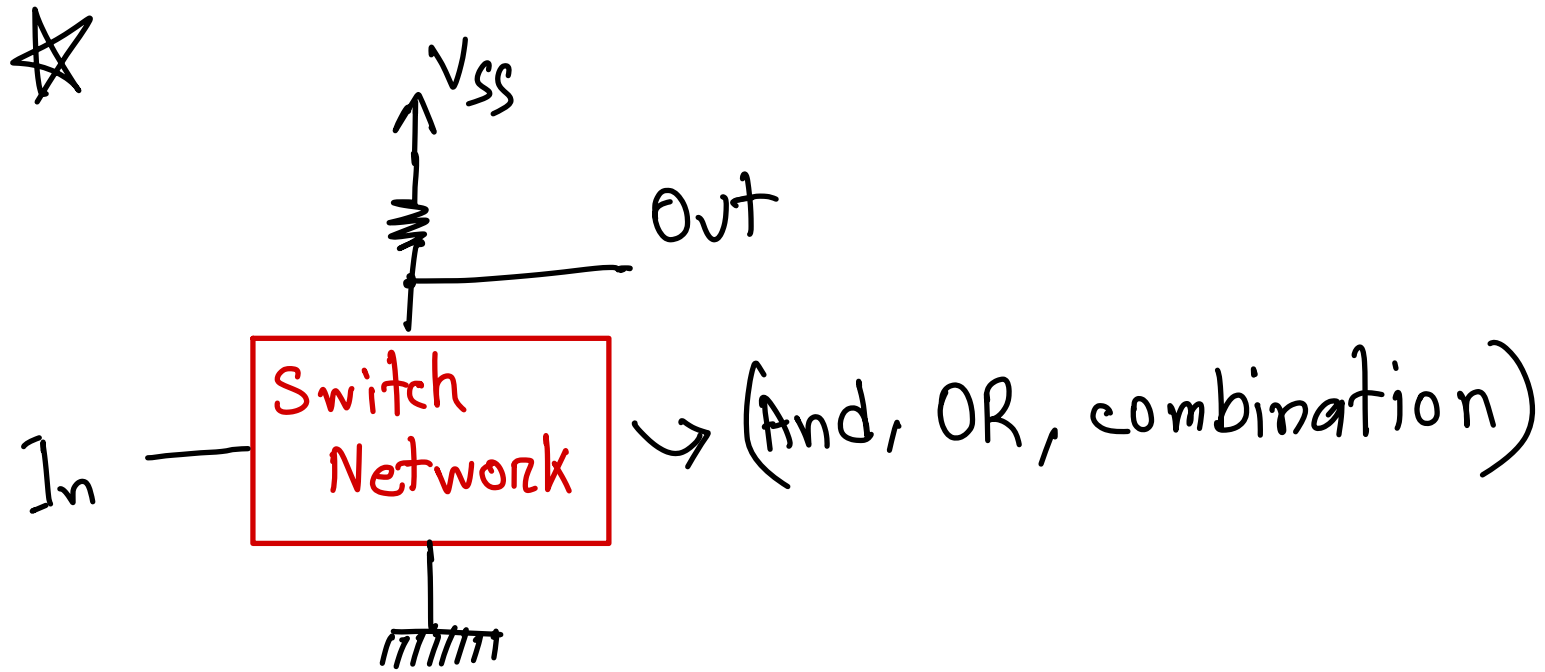
$$OUT = \overline{AB}$$

A	B	V _{OUT}
0	0	5V
0	1	5V
1	0	5V
1	1	0V



$$OUT = \overline{A+B}$$

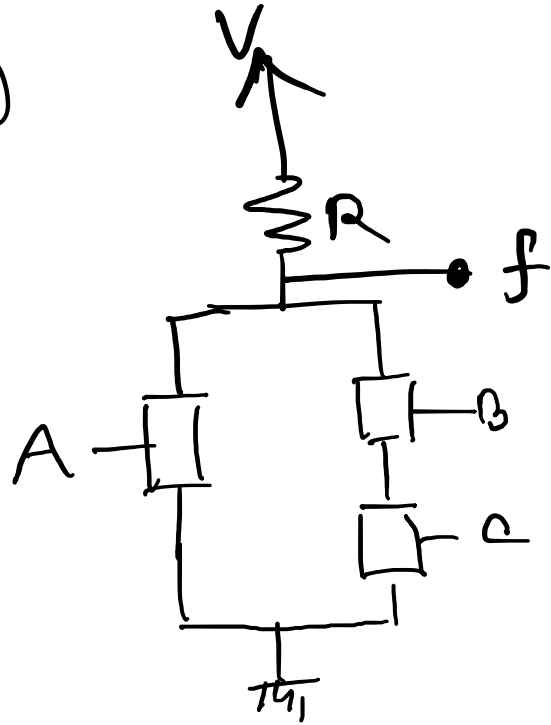
A	B	V _{OUT}
0	0	5V
0	1	0V
1	0	0V
1	1	0V



Switch network ছাড়াও V_{ss} , ground, R থাকে, অর্থাৎ
 বিভিন্ন output **Invert** হয়ে যাবে। AB হওয়া কথা,
 কিন্তু হয়ে যাবে $\overline{AB}$

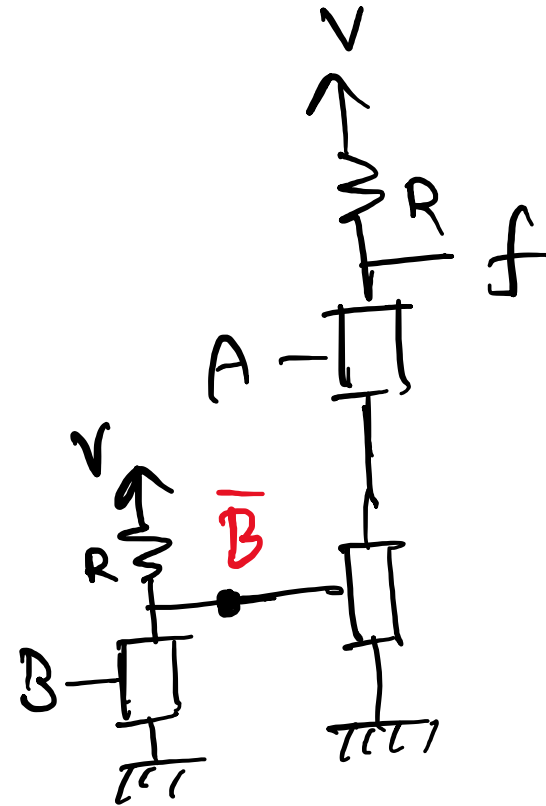
Examples

①



$$f = \overline{A + B.C}$$

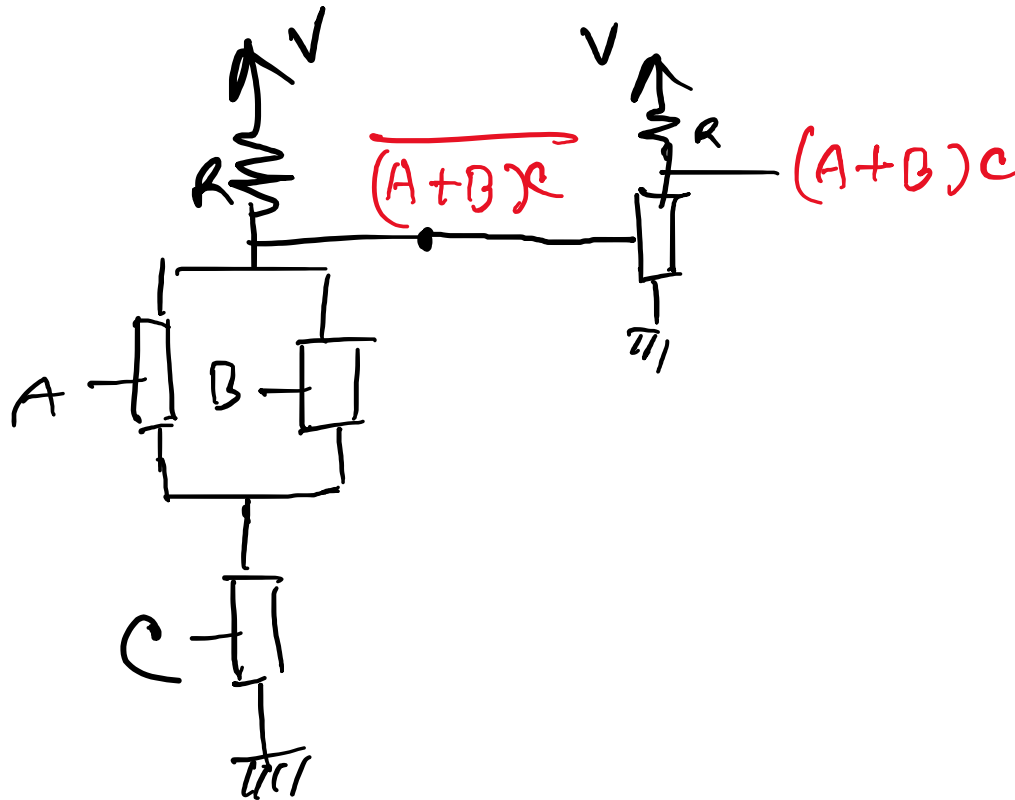
②



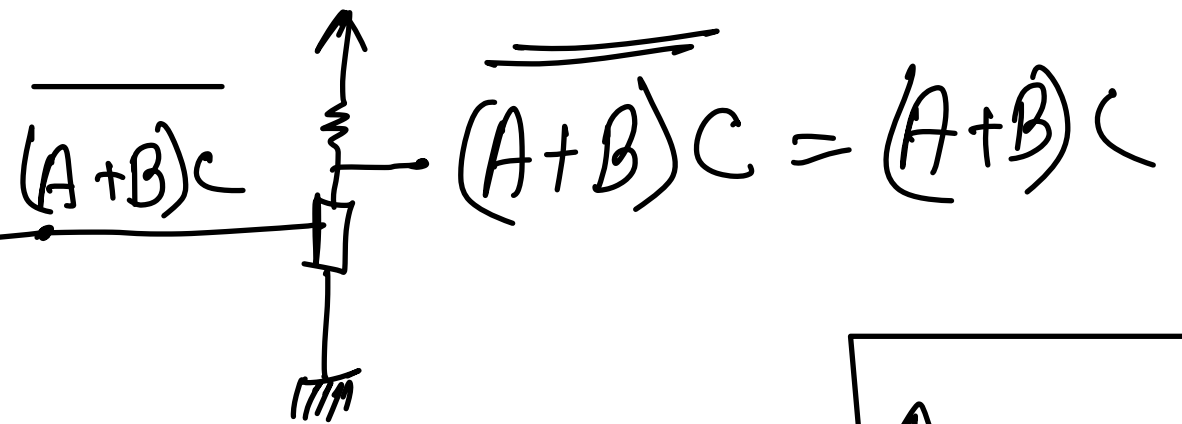
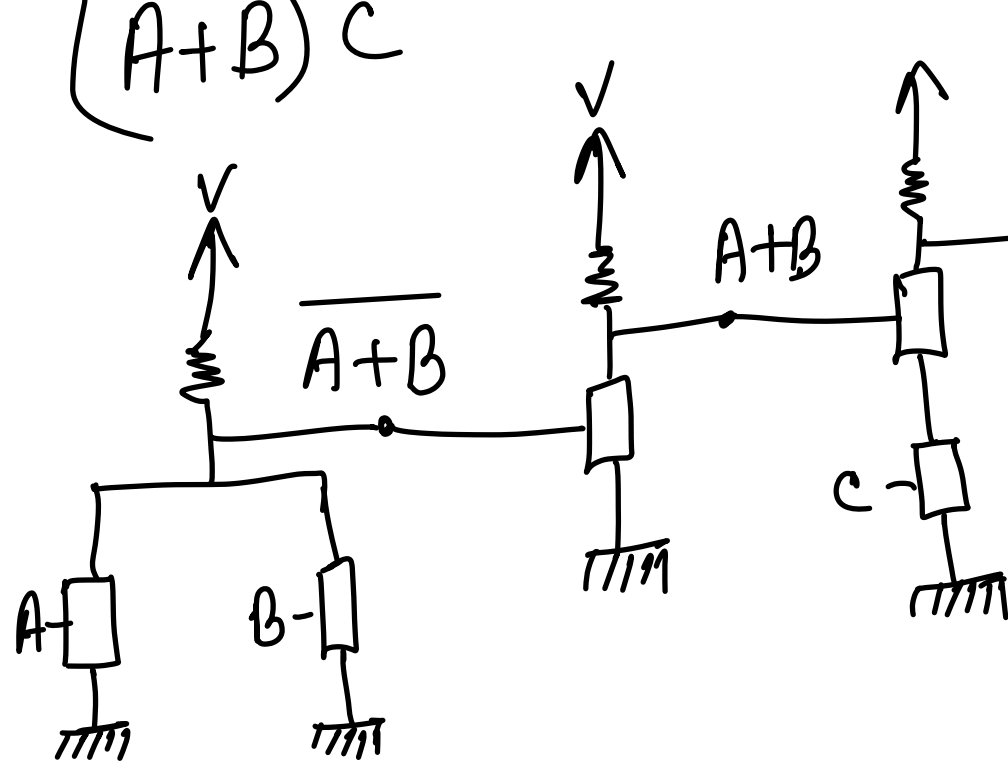
$$f = \overline{A \overline{B}}$$

③ Example

Implement using switches: $f = (A + B)C$

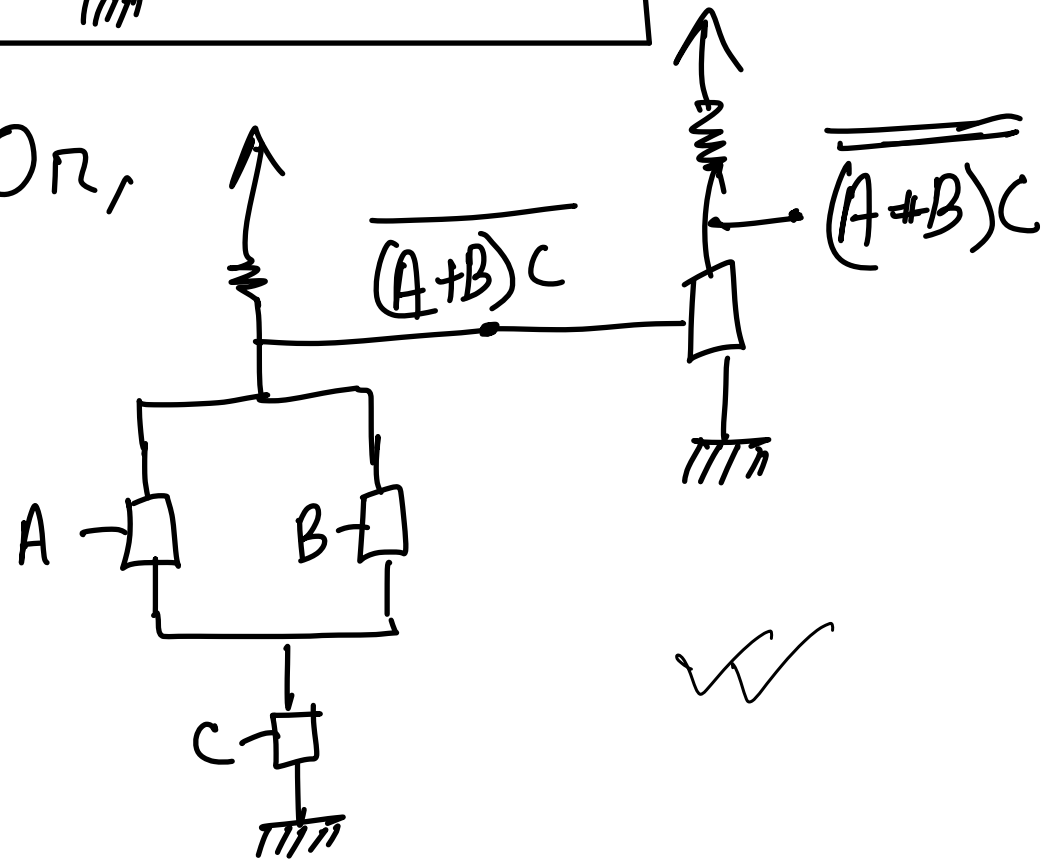


③ $(A+B)C$



$$\overline{\overline{(A+B)C}} = (A+B)C$$

Or,



Always, there will be various ways to draw the circuit.

try to think and find out the most easy way.

Digital Representation

- Binary $\rightarrow$ Two states (0/False, 1/True)
- Binary variables in circuit, need to use two states of device/parameters

Voltage

5V	$\rightarrow$	1
0V	$\rightarrow$	0

0V $\rightarrow$ 1
 3.3V $\rightarrow$ 0

Current

2mA $\rightarrow$ 1
 3mA $\rightarrow$ 0

State

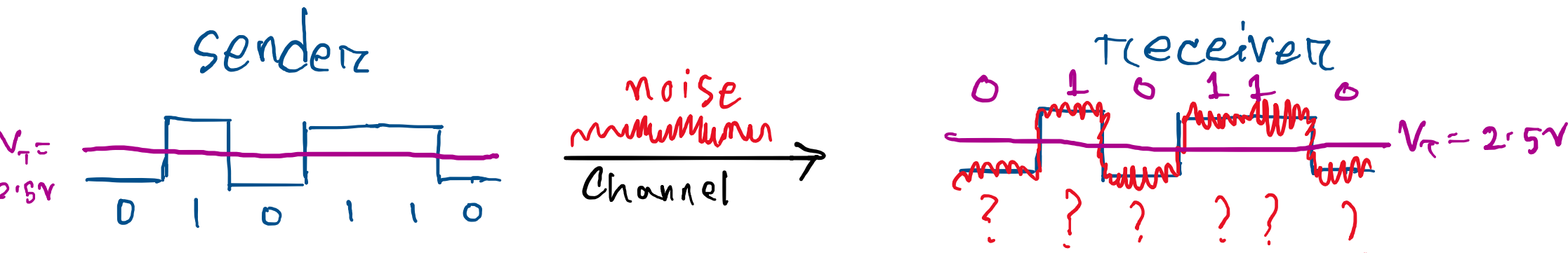
ON $\rightarrow$ 1
 OFF $\rightarrow$ 0

Low resistance $\rightarrow$ 1
 High resistance $\rightarrow$ 0

Skipped

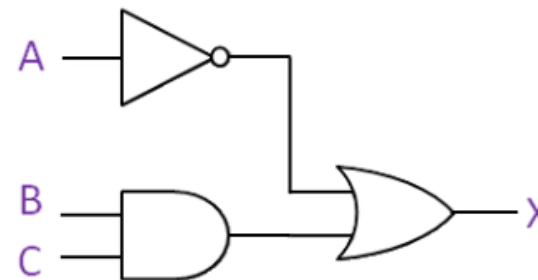
Digital Representation

Suppose you want to send 010110



- Single value based representation fails in the presence of noise
- Better approach – threshold-based system
- Simplest: **Logical 0** = $V < V_T$ **Logical 1** = $V > V_T$

Static Discipline

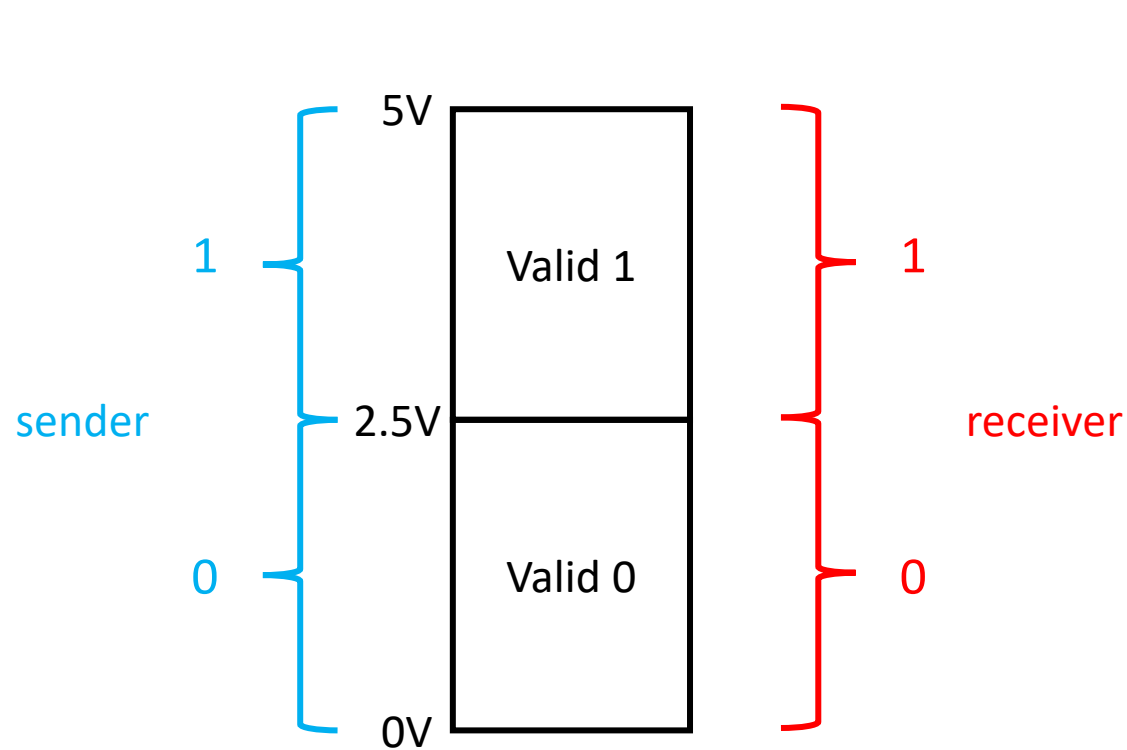


- Specification for digital devices
- Requires devices to adhere to common representation to ensure that **valid input produces valid output**
- This means, if
 - Sender sends "0" $\xrightarrow[\text{Channel}]{\text{noise}}$ Receiver interprets as "0"
 - Sender sends "1" $\xrightarrow[\text{Channel}]{\text{noise}}$ Receiver interprets as "1"

skipped

Static Discipline

Naïve approach: Single threshold based system



$$V_T = 2.5 V$$

sender

Logical 0: $V_o < V_T$
Logical 1: $V_o > V_T$

receiver

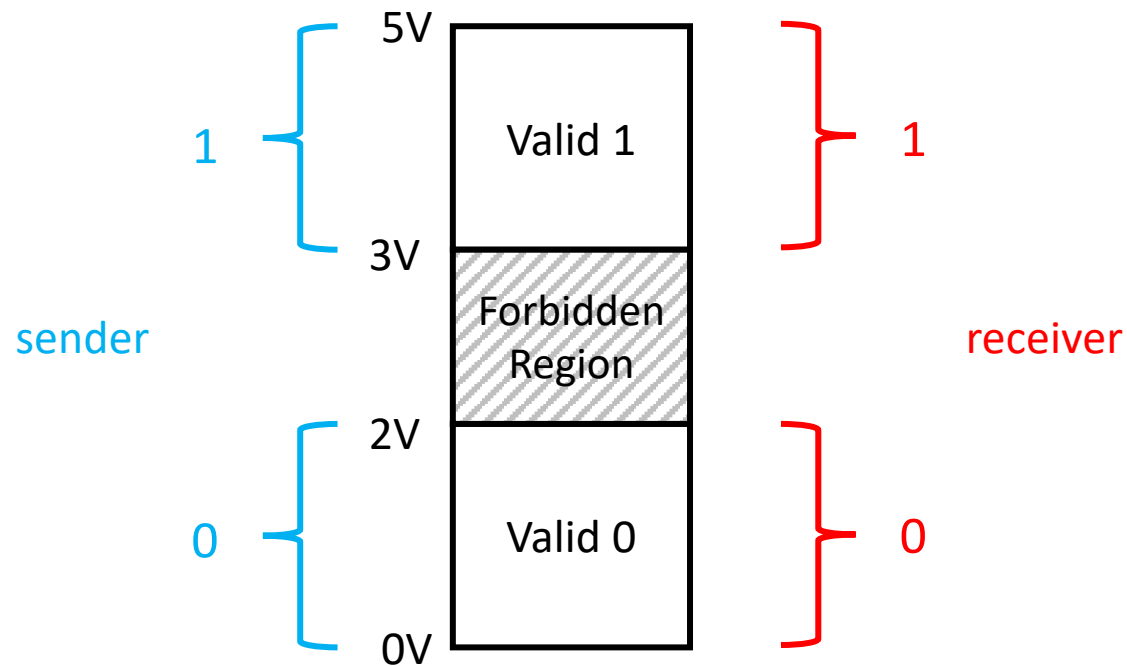
Logical 0: $V_i < V_T$
Logical 1: $V_i > V_T$

What if $V_i = 2.5V$? Undefined!

Skipped

Static Discipline

Double threshold based system



V_H = High voltage threshold = $3V$

V_L = Low voltage threshold = $2V$

sender

Logical 0: $V_o < V_L$
Logical 1: $V_o > V_H$

receiver

Logical 0: $V_i < V_L$
Logical 1: $V_i > V_H$

What if $V_o = 1.9V$ and channel noise is $0.5V$?

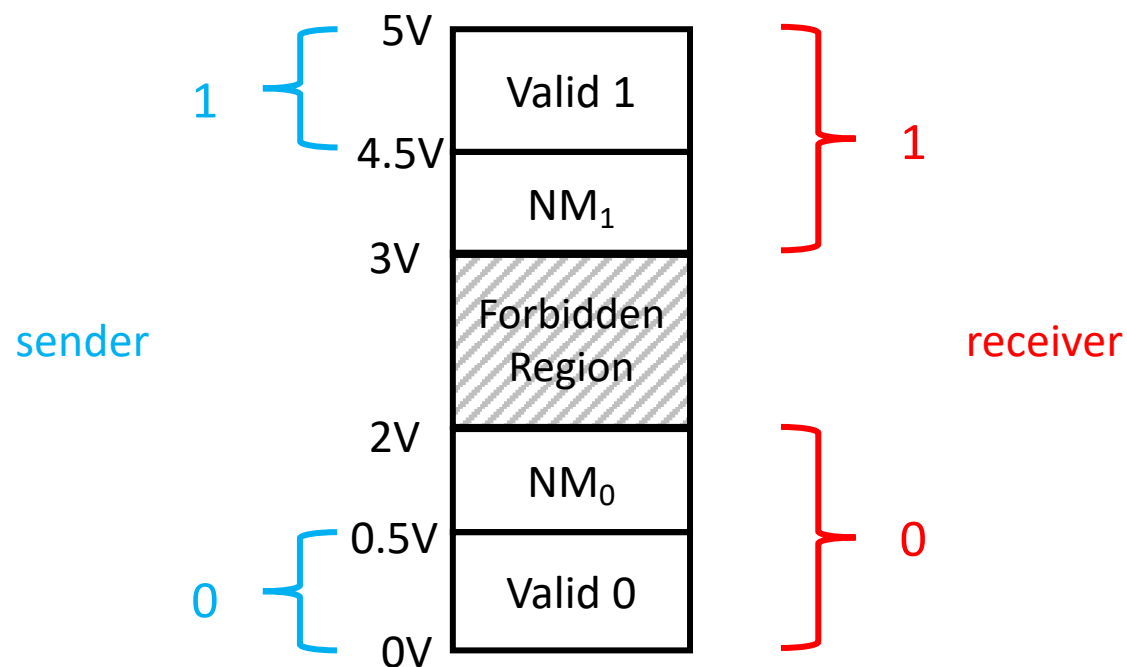
$V_i = 1.9V + 0.5V = 2.4V = \text{invalid}$

➔ valid output producing invalid input,
i.e., no margin for noise

Skipped

Static Discipline

Four threshold based system → Tighter restriction on sender (**output**)



V_{OH} = Output high voltage threshold = 4.5V

V_{OL} = Output low voltage threshold = 0.5V

V_{IH} = Input high voltage threshold = 3V

V_{IL} = Input low voltage threshold = 2V

sender

Logical 0: $V_o < V_{OL}$
Logical 1: $V_o > V_{OH}$

receiver

Logical 0: $V_i < V_{IL}$
Logical 1: $V_i > V_{IH}$

For static discipline, $V_{OL} < V_{IL} < V_{IH} < V_{OH}$

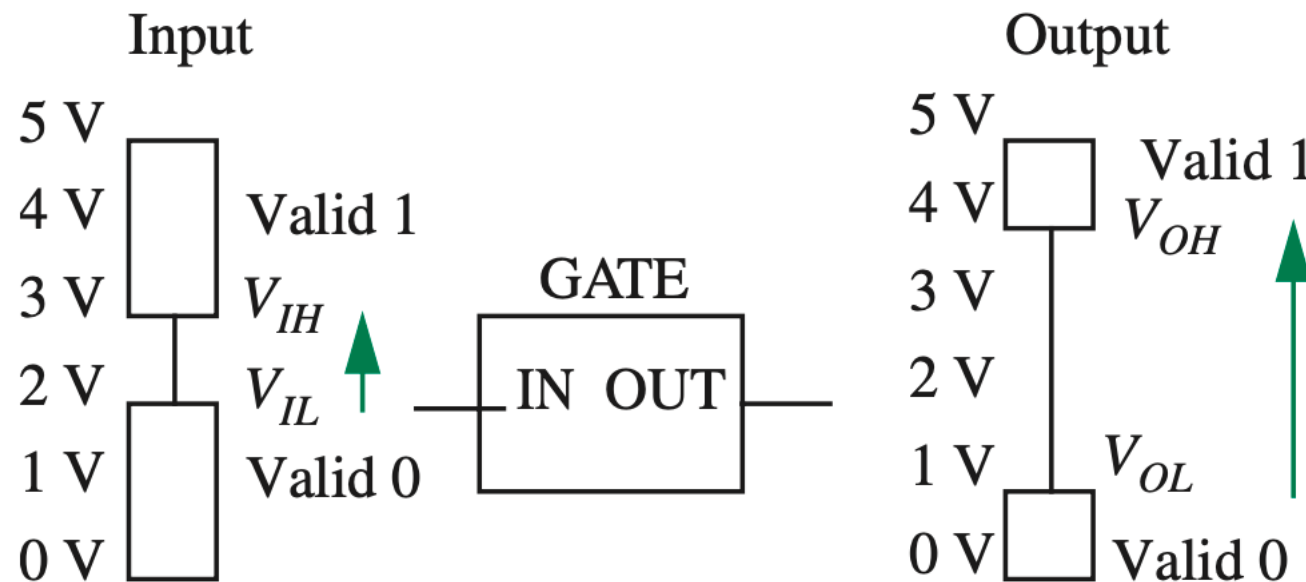
Noise margins (NM):

- $NM_1 = V_{OH} - V_{IH} = 4.5 - 3 = 1.5V$ (significance?)
- $NM_0 = V_{IL} - V_{OL} = 2 - 0.5 = 1.5V$ (significance?)

skipped

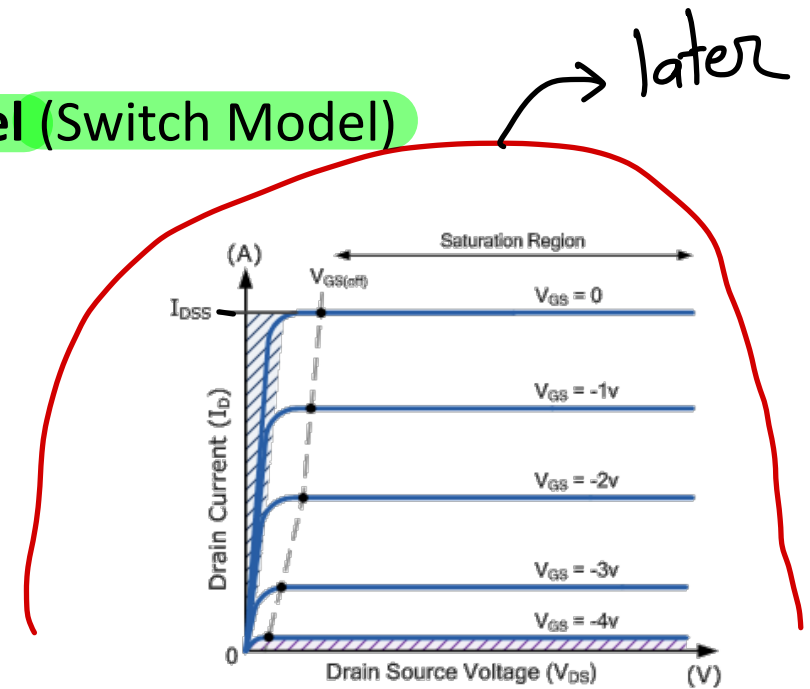
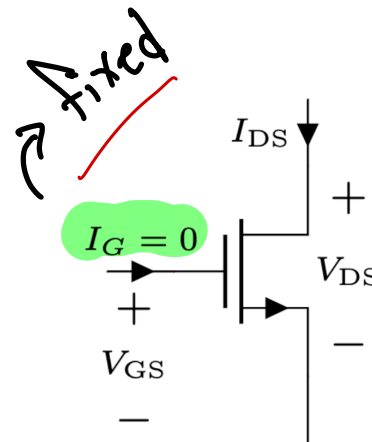
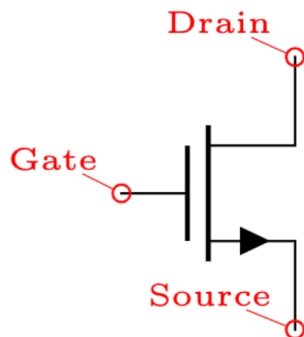
Static Discipline

Four threshold based system → Tighter restriction on sender (**output**)



Transistors as Digital Switch

- Transistors are 3 terminal non-linear devices, can be used as switch
- 2 types – Voltage Controlled, Current Controlled
- **Metal Oxide Semiconductor Field Effect Transistor (MOSFET)** are voltage controlled
- Control, $C = V_{GS}$. The IV characteristics (I_{DS} vs V_{DS}) depends on V_{GS}
- Actual dependency is complex.
- Will start with a simple (but approximate) one – **S-Model (Switch Model)**



→ Switch

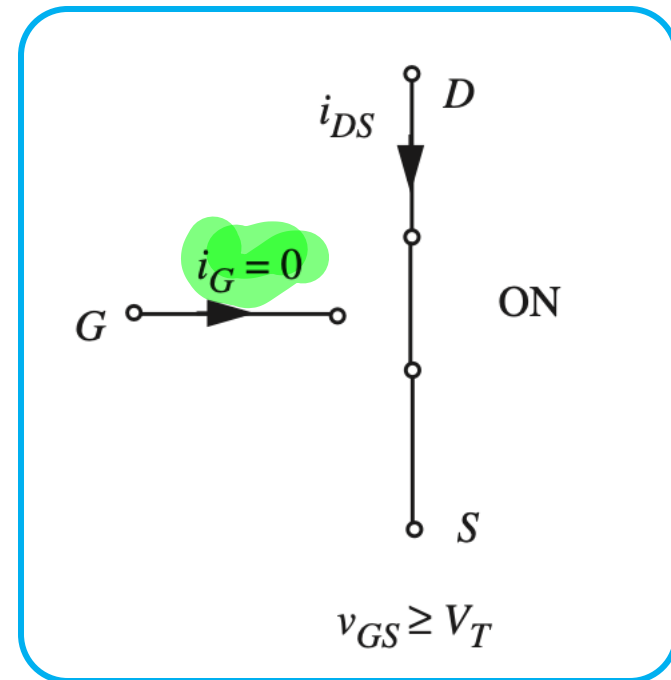
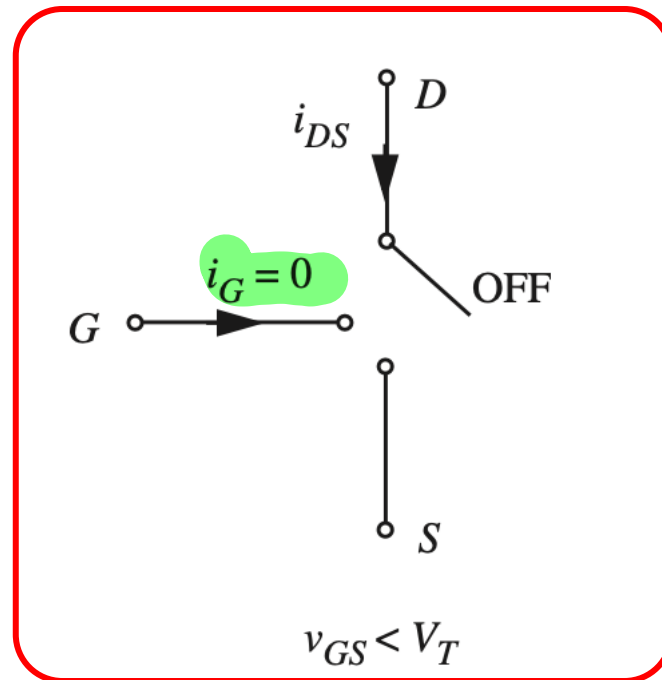
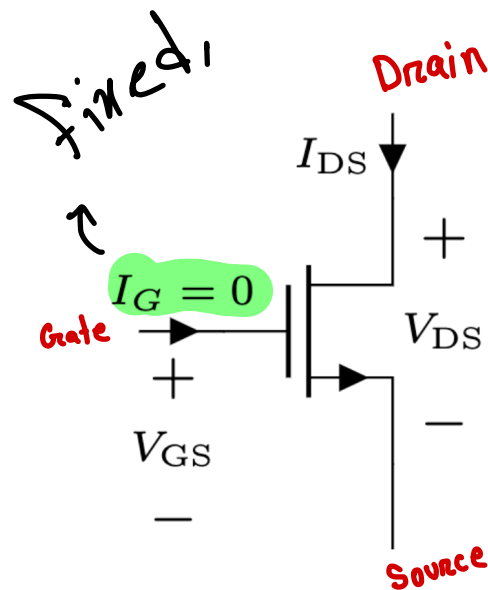
MOSFET S-Model

Control terminal = Gate

V_T = threshold voltage

- The MOSFET (approximately) behaves like a switch

- $C = V_{GS}$. Here, $C = "0" \Rightarrow V_{GS} < V_T$, and $C = "1" \Rightarrow V_{GS} \geq V_T$

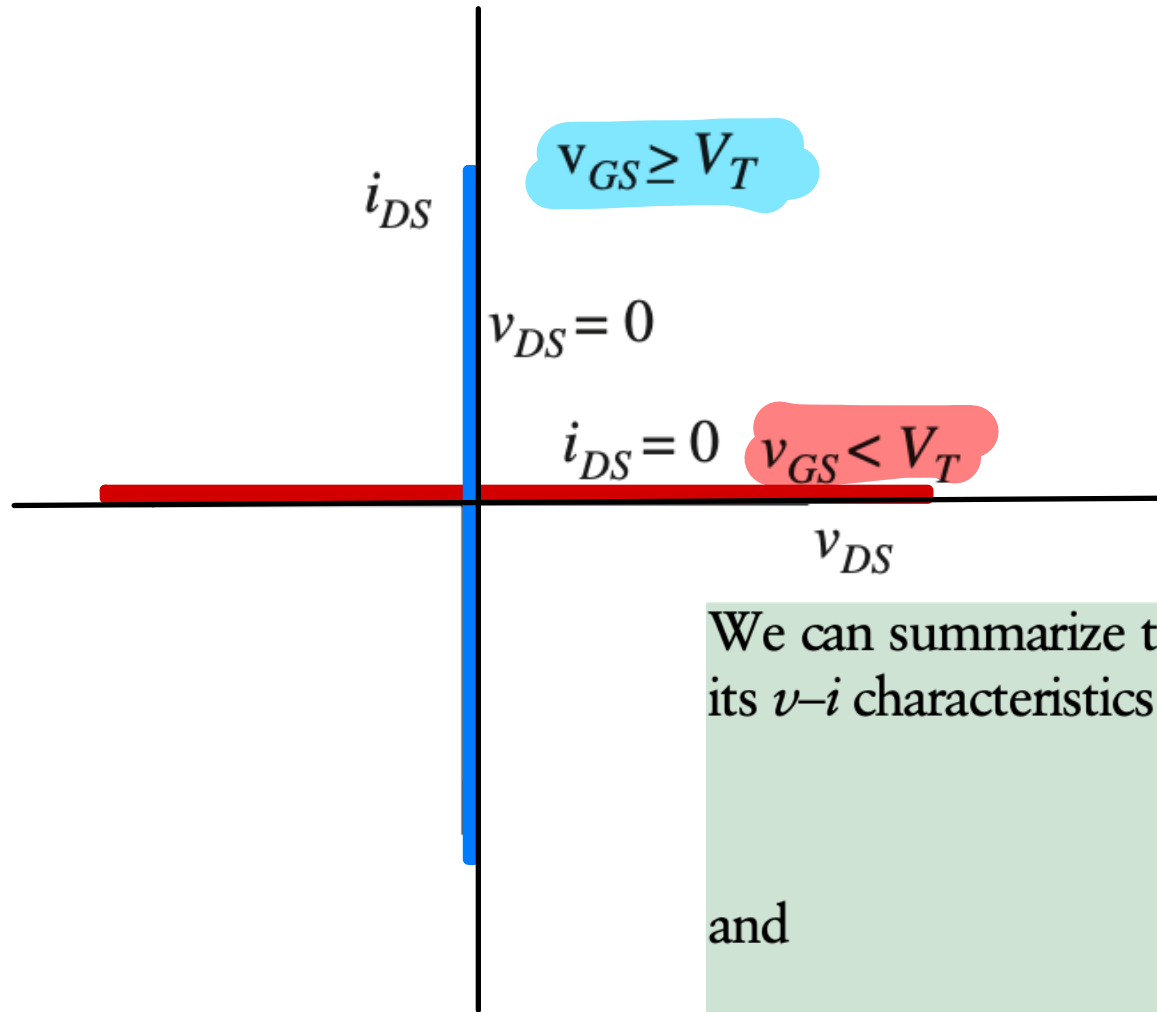


V_{GS}

-	+
$c=0$	$c=1$
V_T	

$V_{GS} [V]$

MOSFET S-Model



We can summarize the S model for the MOSFET in algebraic form by stating its v - i characteristics as follows:

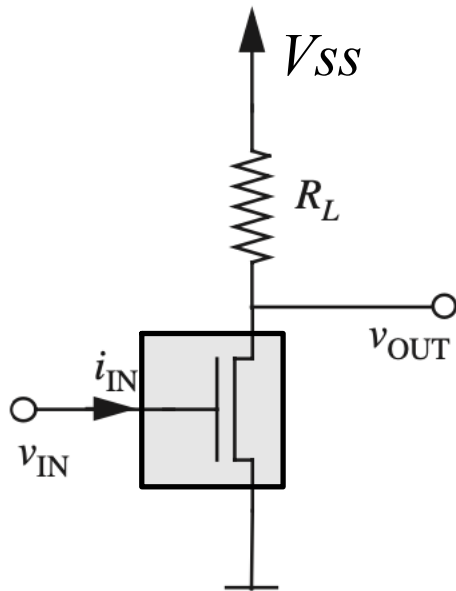
$$\text{for } v_{GS} < V_T, \quad i_{DS} = 0$$

and

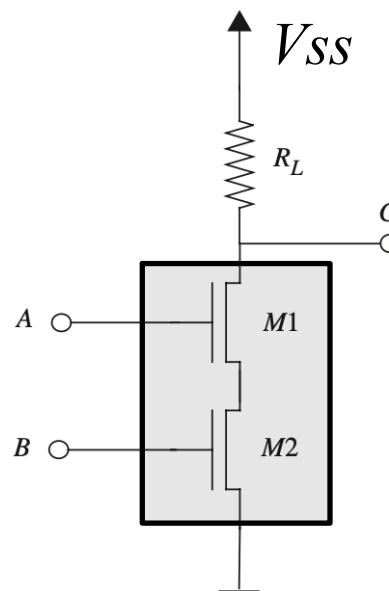
$$\text{for } v_{GS} \geq V_T, \quad v_{DS} = 0 \quad (6.2)$$

Logic Gates using MOSFET

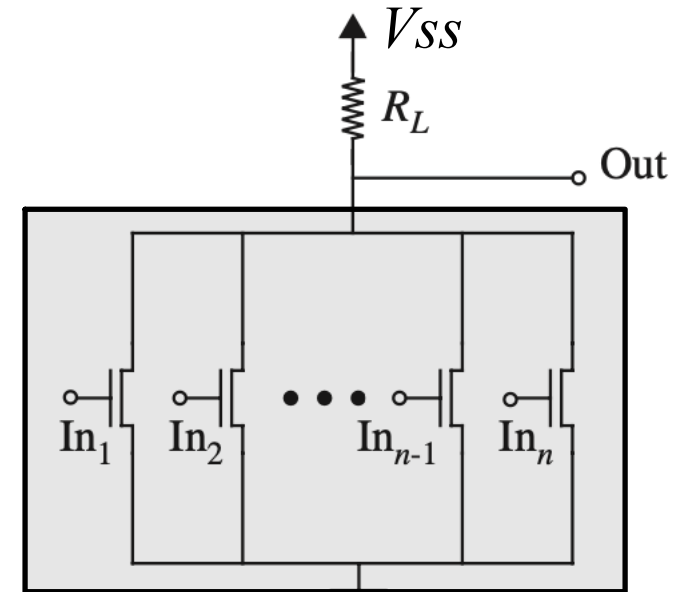
Just replace the switches with MOSFETs!



NOT Gate (Inverter)

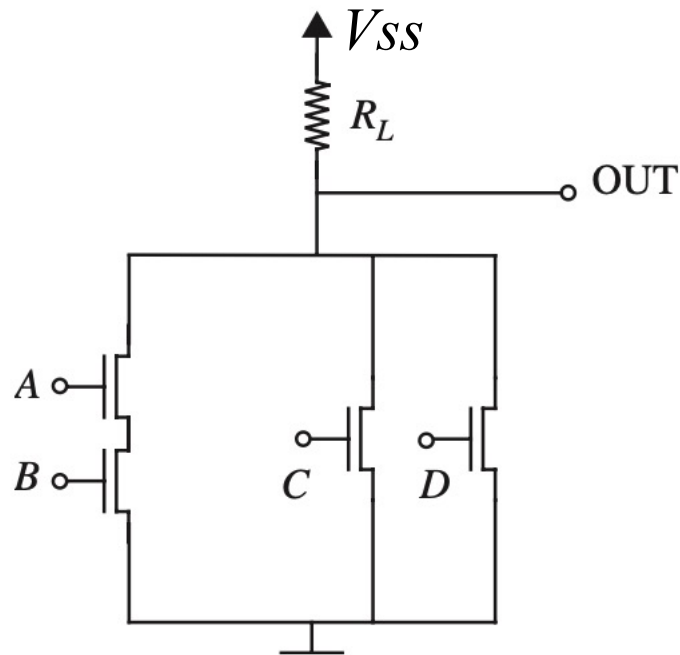


NAND Gate (Inverter)

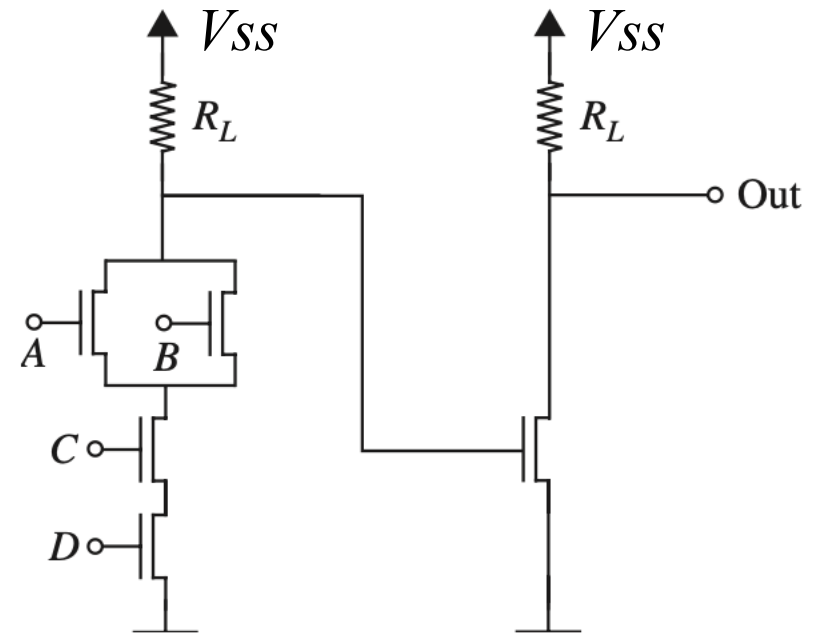


NOR Gate (Inverter)

MOSFET Logic Gates – More Examples



$$OUT = \overline{AB + C + D}$$

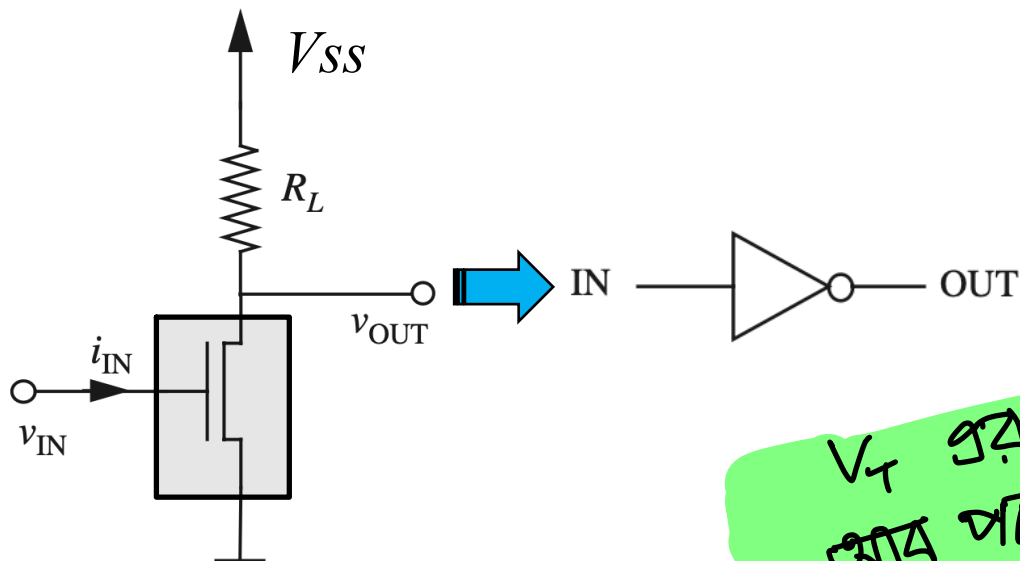


$$Out = \overline{\overline{(A + B)CD}} = (A + B)CD$$

→ Output vs input

Voltage Transfer Characteristics (VTC) = (v_{OUT} vs v_{IN})

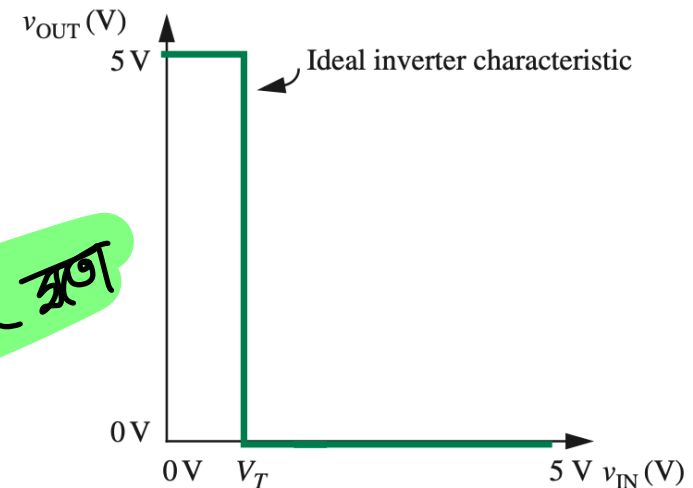
- Reminder: VTC is a graph where x axis = input voltage, y axis = output voltage
- Why? Design logic gates to follow a given static discipline



MOSFET Auto NOT gate

When $v_{IN} < V_T$ (Logical 0), $v_{OUT} = V_S = 5V$ (Logical 1)

When $v_{IN} \geq V_T$ (Logical 1), $v_{OUT} = 0$ (Logical 0)



V_T এর আর
আর পরে inverter করা
হয়ে থাকে

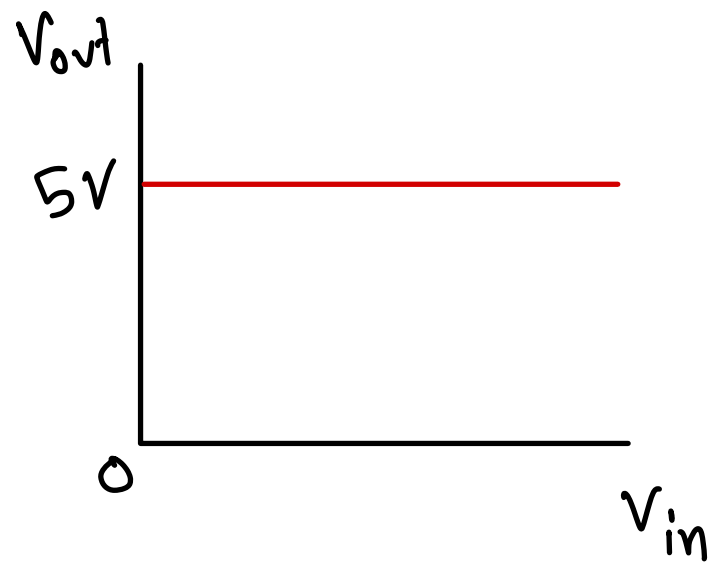
When,

$$V_{in} < V_T$$

$$\Rightarrow A = \text{off} = 0$$

then,

$$\therefore V_{out} = V_{SS} = 1$$

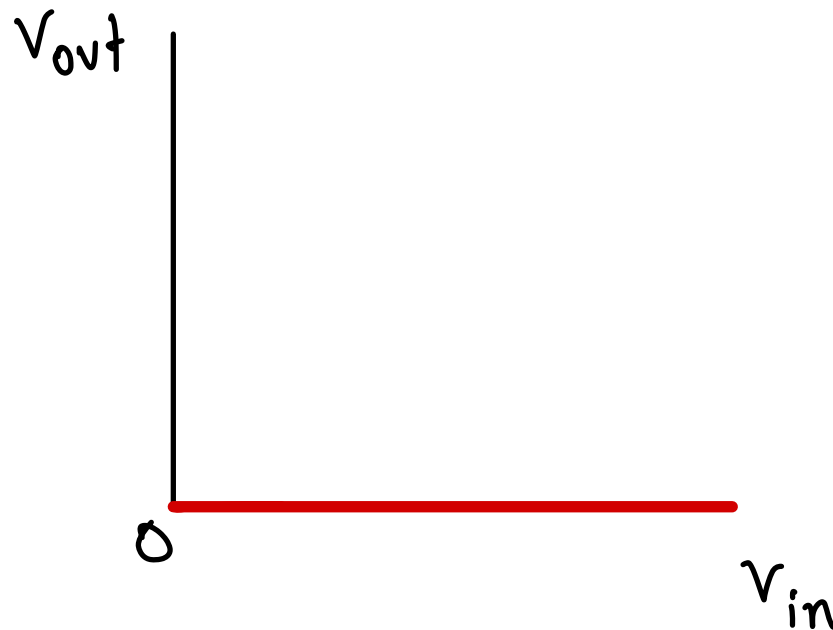


when,

$$V_{in} > V_T$$

$$\Rightarrow A = \text{on} = 1$$

$$\therefore \text{then, } V_{out} = \text{ground} = 0$$



Input 0 शून्य

output 1 एक

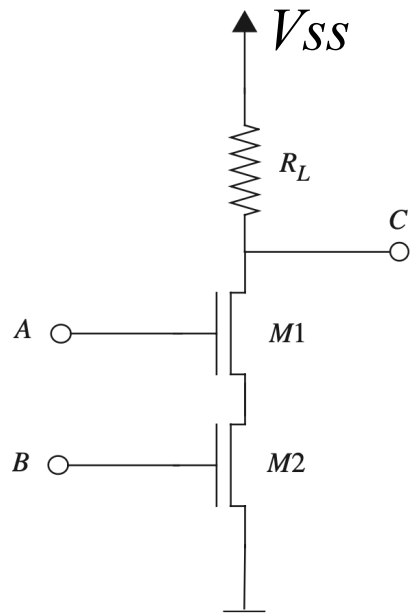
vice versa

It's working
like an inverter

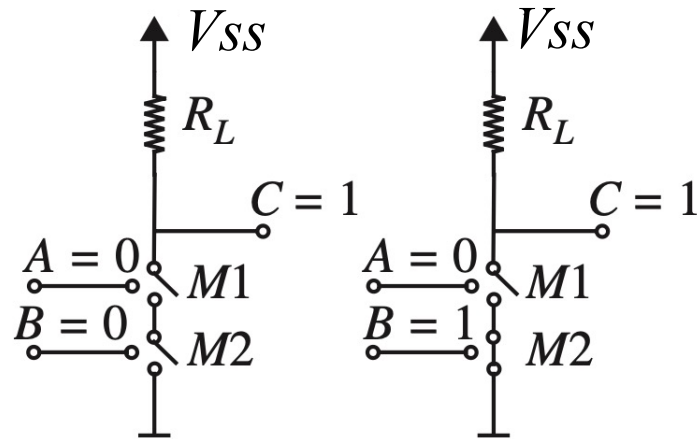
here's the
reference
point is
threshold.

VTC of NAND gate

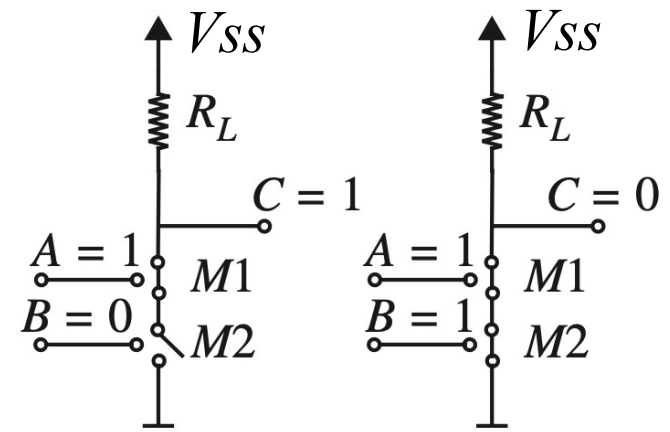
- We only have one x axis, but two **inputs** *variable*
- Solution: **Draw two VTC**, one considering **$A = "0"$** , one considering **$A = "1"$**

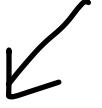


When $A = "0"$



When $A = "1"$

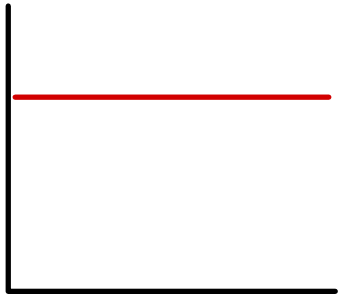




$$A = 0$$

$$B = 0$$

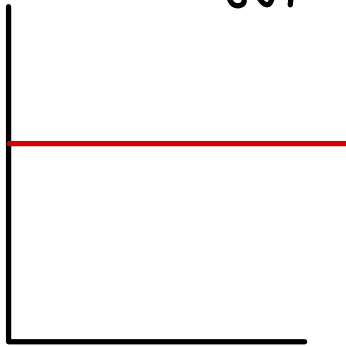
$$V_{out} = 1$$



$$A = 0$$

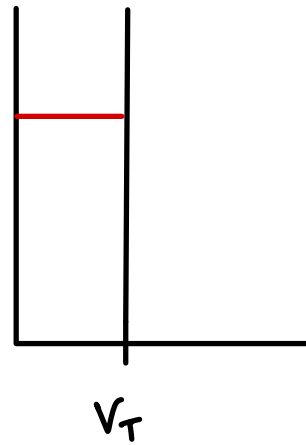
$$B = 1$$

$$V_{out} = 1$$



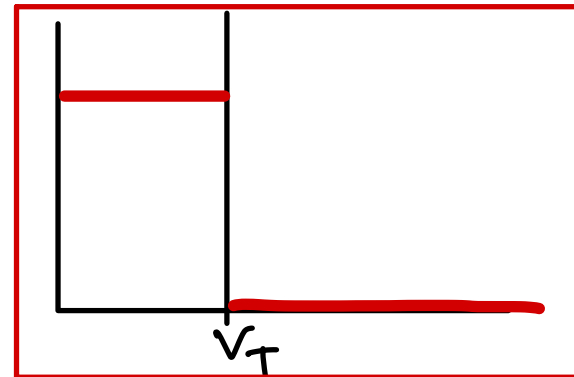
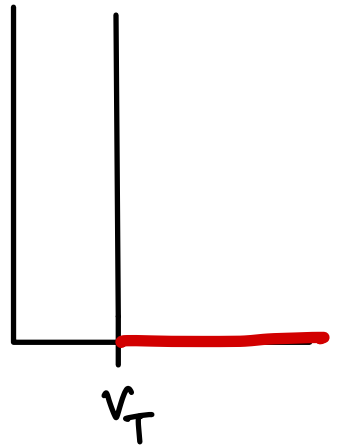
$$A = 1$$

$$B = 0$$



$$A = 1$$

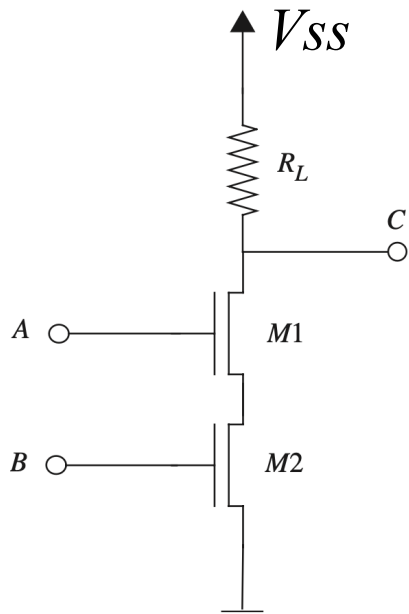
$$B = 1$$



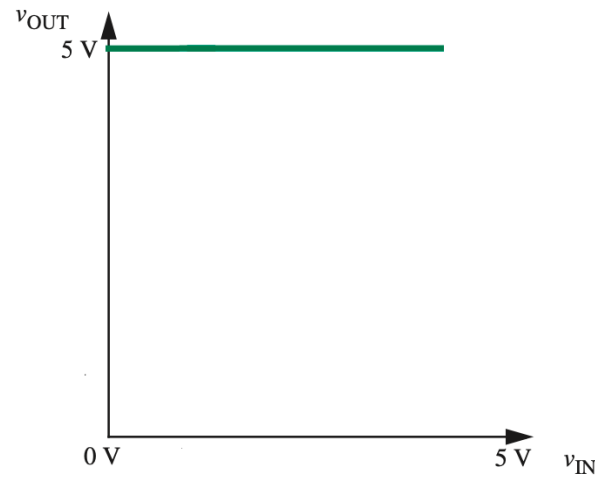
→ direct draw this

VTC of NAND gate

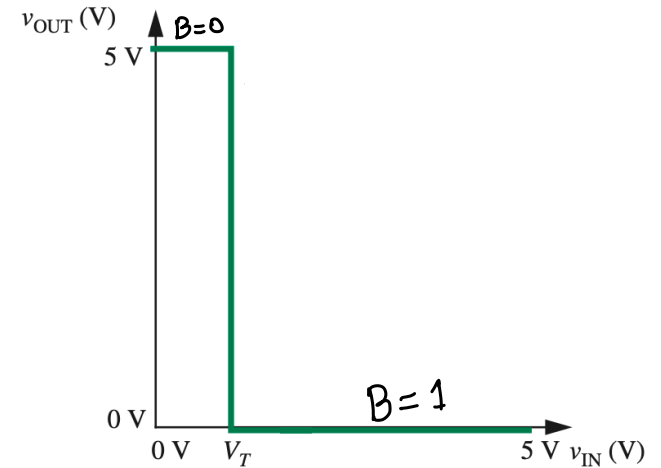
- We only have one x axis, but two inputs
- Solution: Draw two VTC, one considering $A = "0"$, one considering $A = "1"$



When $A = "0"$

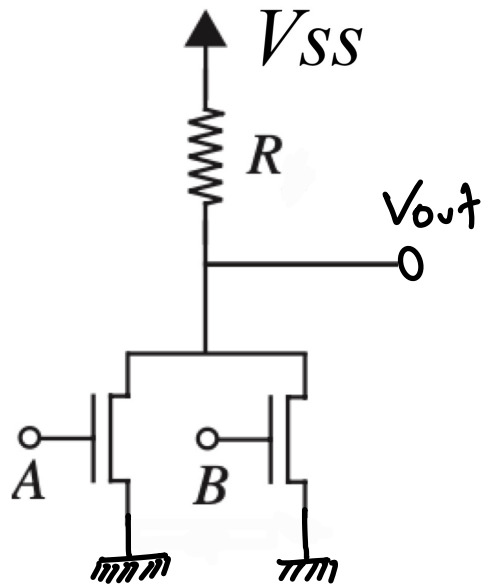


When $A = "1"$



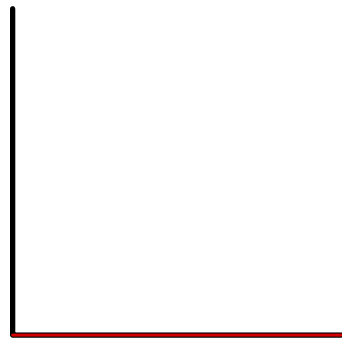
Homework: Find VTC for NOR gate

H.W → VTC for NOR Gate



$$A = 1$$

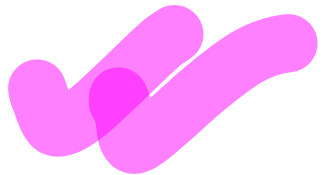
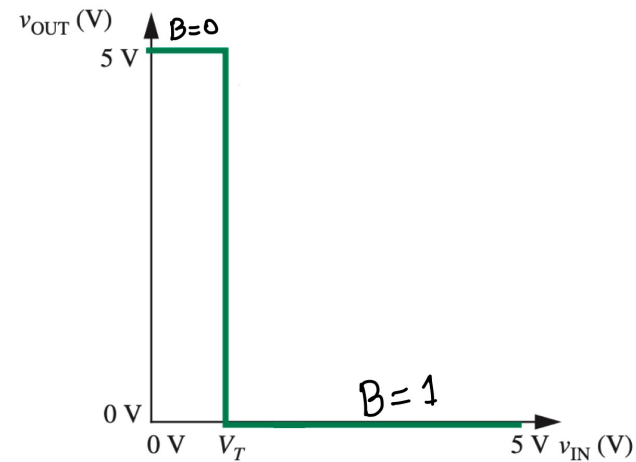
$$\left. \begin{array}{l} B = 0 \\ B = 1 \end{array} \right\} V_{out} = 0$$



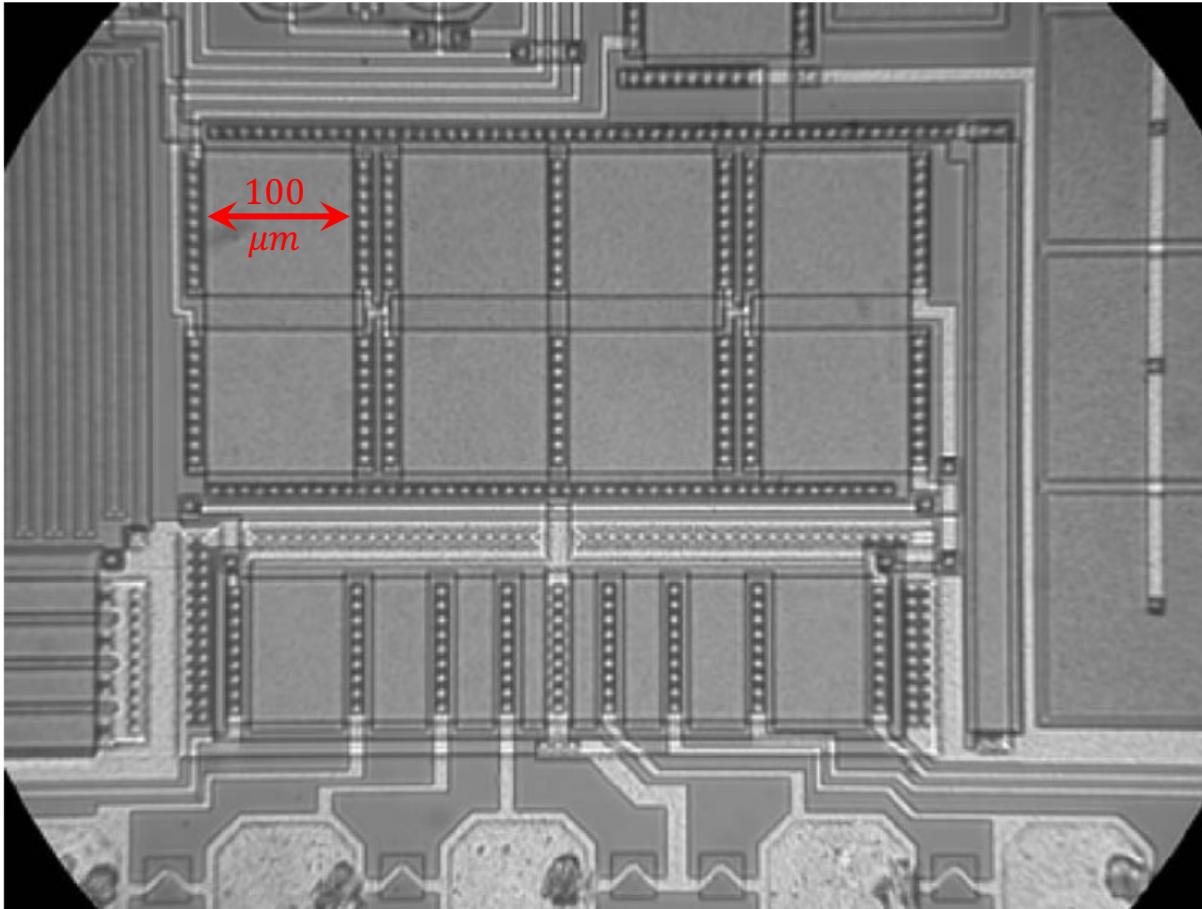
$$A = 0$$

$$B = 0 \rightarrow V_{out} = V_{SS}$$

$$B = 1 \rightarrow V_{out} = 0$$



Construction of Real MOSFET

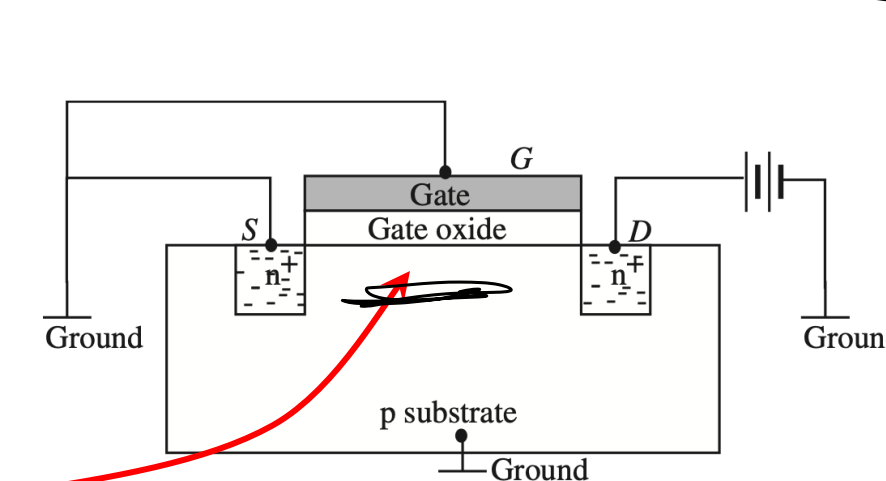
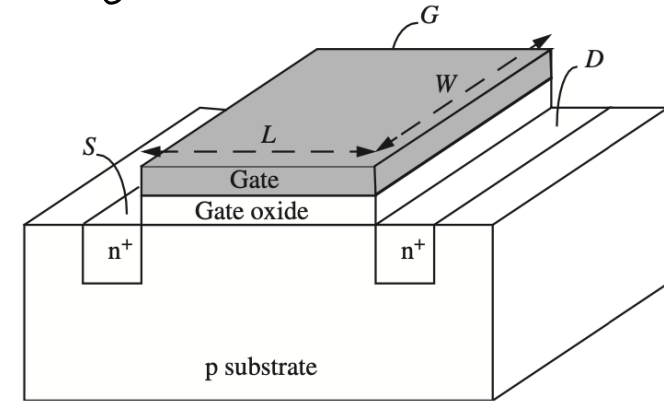
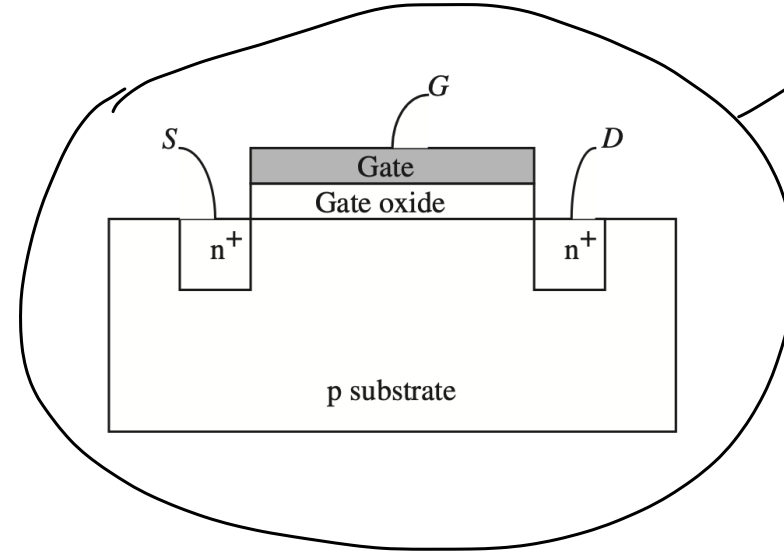


Top view of several n-channel MOSFETs fabricated on a chip. The square MOSFETs in the center of the photograph have a width and length of 100 μm. (Photograph Courtesy of Maxim Integrated Products.)

Construction of Real MOSFET

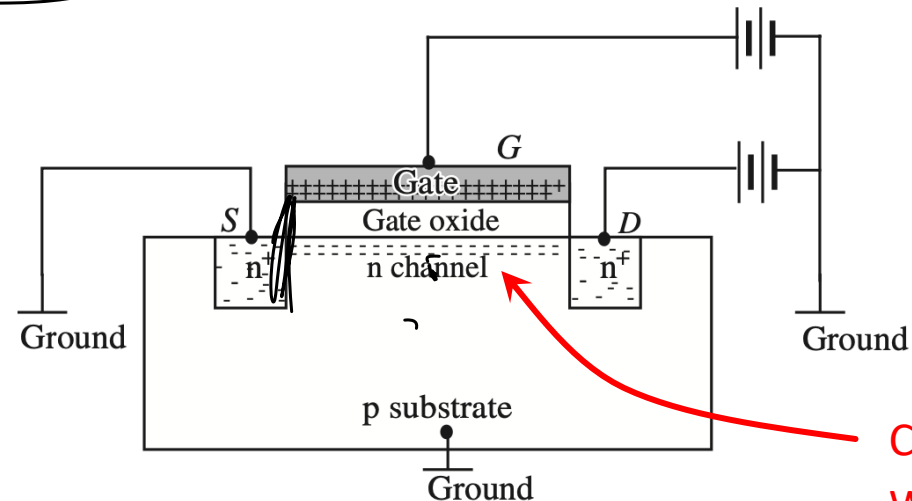
n type MOSFET

Simplified cross section and 3D view



No channel, open ckt

$$V_{GS} = 0V$$



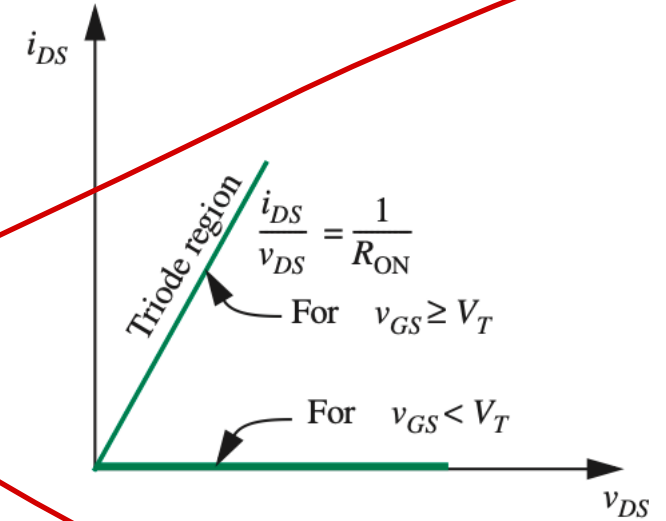
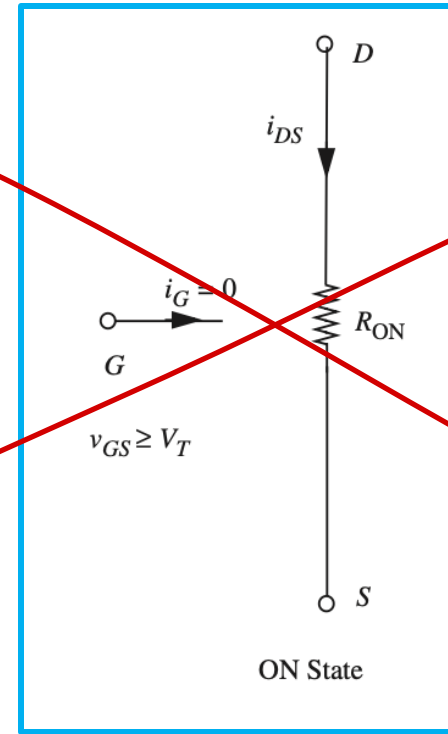
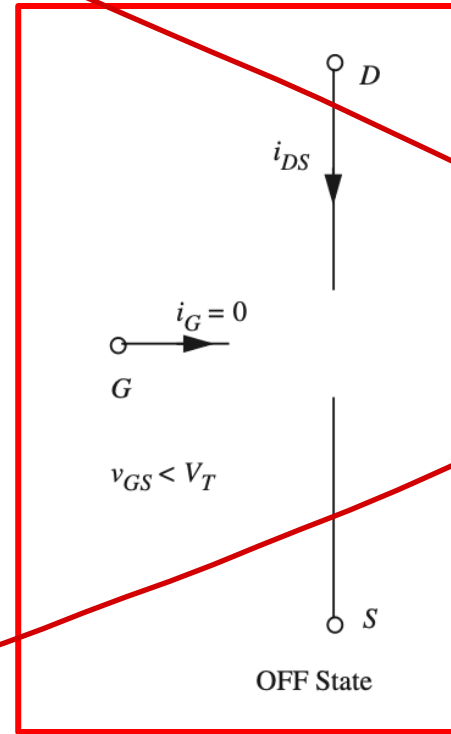
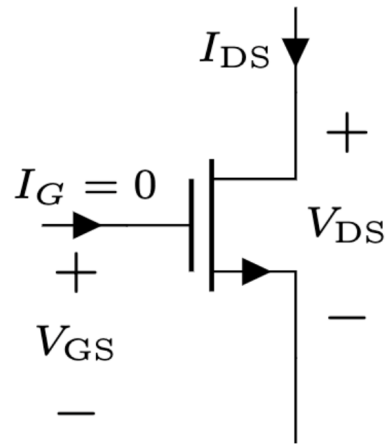
Channel created
Will have some R
-> SR model

$$V_{GS} > V_T$$

~~SR Model~~

Excluded in

Summer 24



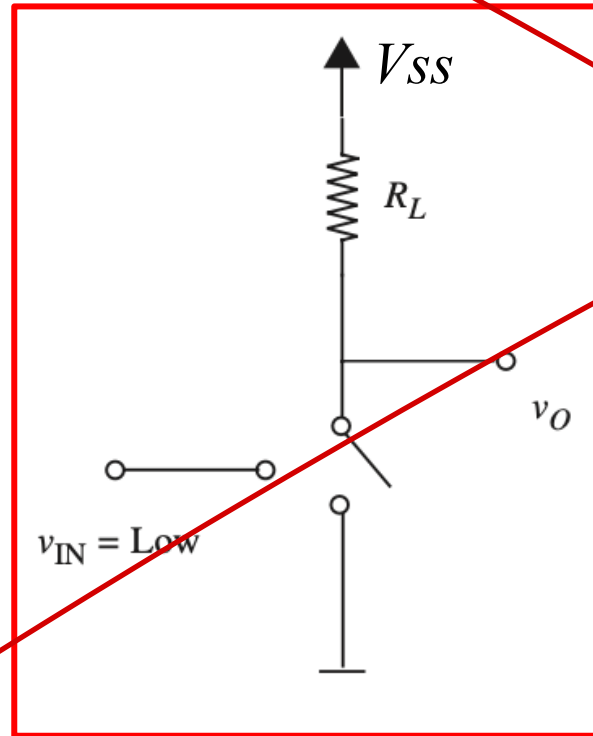
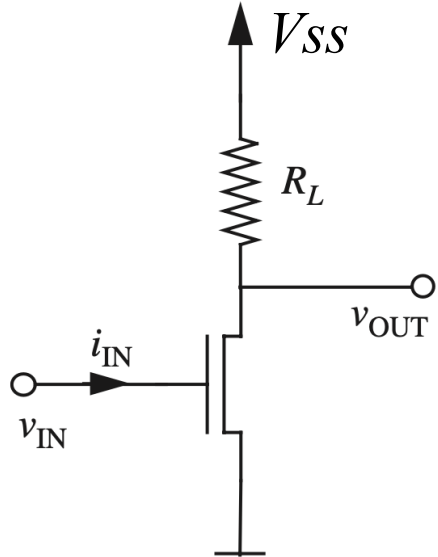
$$R_{ON} = \frac{1}{k'_n \frac{W}{L} (V_{GS} - V_T)} = \frac{1}{k V_{OV}}$$

Unit of $k = \text{mA}/V^2$

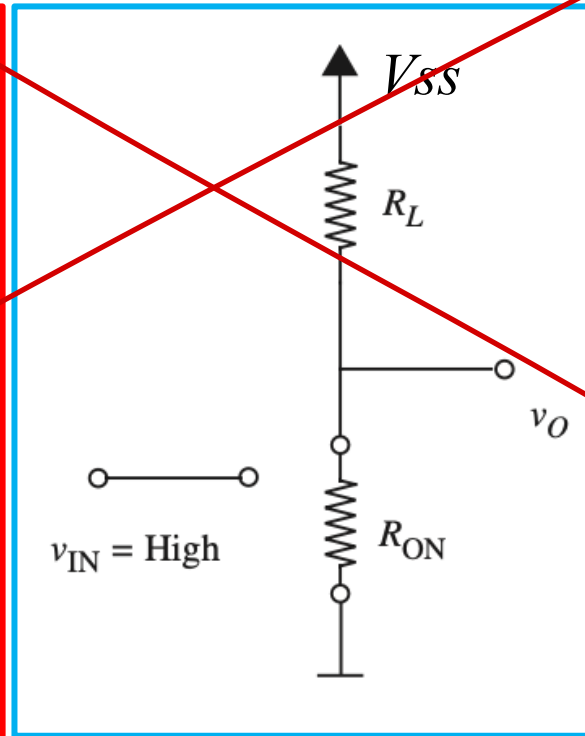
- SR model is a better approximation than S model
- However, still an approximation. This model fails when V_{DS} increases to around $V_{GS} - V_T$

SR Model - Inverter

Excluded



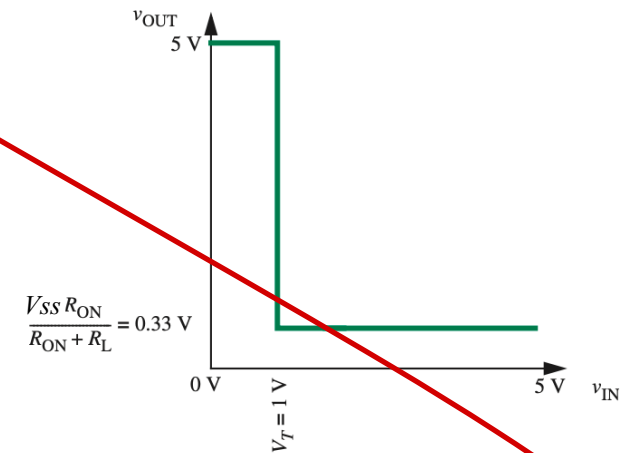
$$v_{OUT,High} = V_{SS}$$



$$v_{OUT,Low} = V_{SS} \frac{R_{ON}}{R_{ON} + R_L} \neq 0$$

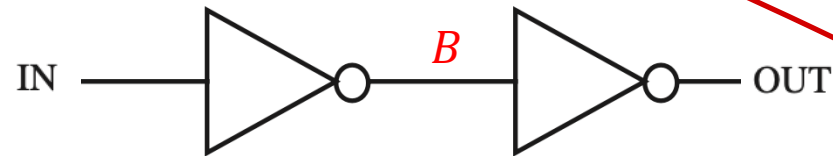
For example, if $V_{SS} = 5V$,
 $R_{ON} = 1\text{ k}\Omega$, $R_L = 14\text{ k}\Omega$

$$V_{SS} \frac{R_{ON}}{R_{ON} + R_L} = 0.33\text{ V.}$$



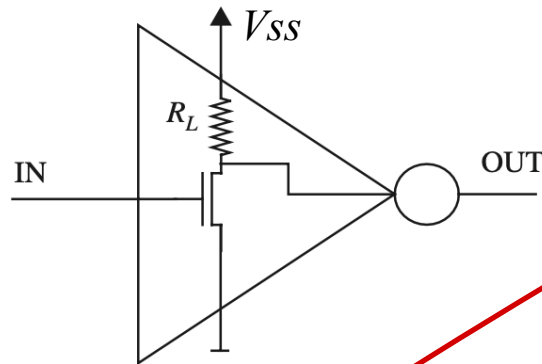
Design of logic gates

Excluded



Expected

IN	B	OUT
Low ($V_{IN} < V_T$)	High (5V)	Low ($V_{OUT} = V_{OUT,Low}$)
High ($V_{IN} > V_T$)	Low ($V_{OUT,Low}$)	High (5V)



Actual

IN	B	OUT
Low ($V_{IN} < V_T$)	High (5V)	Low ($V_{OUT} = V_{OUT,Low}$)
High ($V_{IN} > V_T$)	Low ($V_{OUT,Low} = 0.5 V$)	Low ($V_{OUT} = V_{OUT,Low}$)

$$V_T = 0.4 V, V_{SS} = 5V, R_{ON} = 1 k\Omega, R_L = 9 k\Omega$$

$$V_{OUT,High} = V_{SS} = 5V$$

$$V_{OUT,Low} = V_{SS} \frac{R_{ON}}{R_{ON} + R_L} = 5 \frac{1}{1 + 9} = 0.5 V$$

Therefore, need to design logic gates properly such that

$$V_{SS} \frac{R_{ON}}{R_{ON} + R_L} < V_T.$$

Design of logic gates - Example

Excluded

Assume the following values for the inverter circuit parameters: $V_S = 5\text{ V}$, $V_T = 1\text{ V}$, and $R_L = 10\text{ k}\Omega$. Assume, further, that $\frac{1}{k'_n V_{OV}} = 5$ for the MOSFET. Determine a $\frac{W}{L}$ sizing for the MOSFET so that the inverter gate output for a logical 0 is able to switch OFF the MOSFET of another inverter.

Solution:

$$V_S \frac{R_{ON}}{R_{ON} + R_L} < V_T$$

$$\Rightarrow 5 \frac{R_{ON}}{R_{ON} + 10} < 1$$

$$\Rightarrow 5R_{ON} < R_{ON} + 10$$

$$\Rightarrow R_{ON} < \frac{10}{4} = 2.5$$

Now,

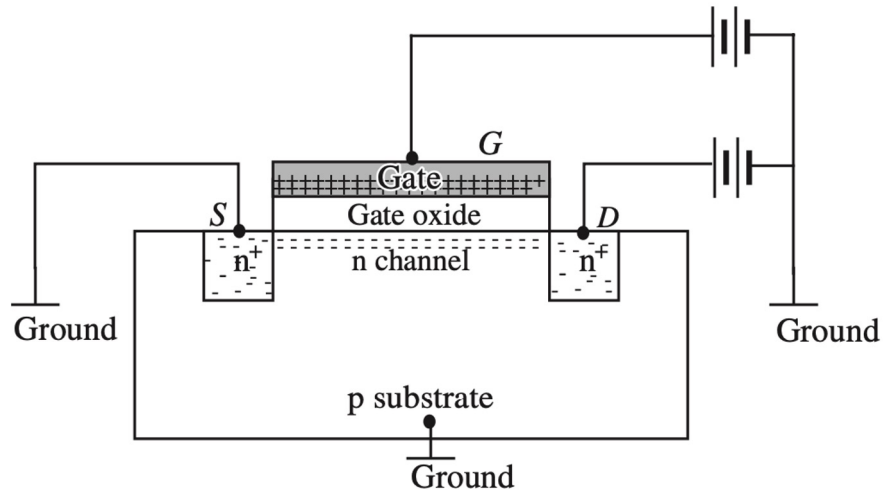
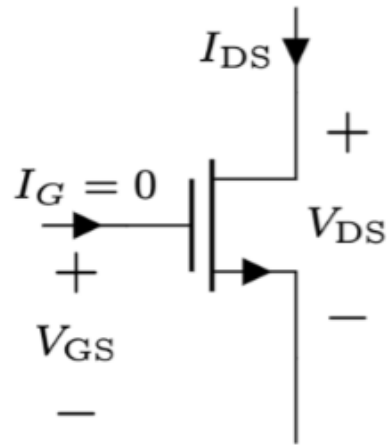
$$R_{ON} = \frac{1}{k'_n \frac{W}{L} V_{OV}} = 5 \times \frac{1}{W/L}$$

Hence

$$\frac{5}{W/L} < 2.5 \Rightarrow \frac{W}{L} > \frac{5}{2.5}$$

$$\Rightarrow \frac{W}{L} > 2$$

Review – MOSFET



Control = $V_{GS} = V_G - V_S$, controls the IV between drain-source (I_{DS} vs V_{DS})

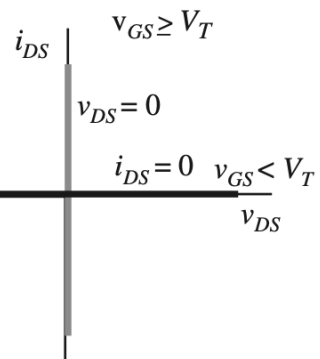
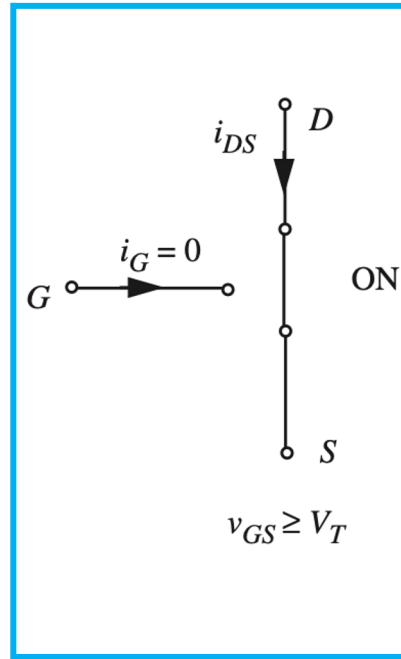
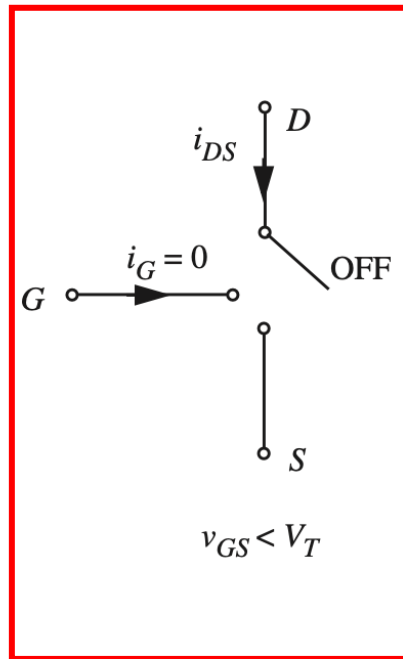
Threshold voltage = V_T , minimum voltage required to create the channel

Models

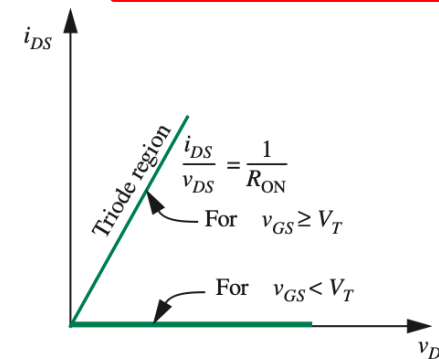
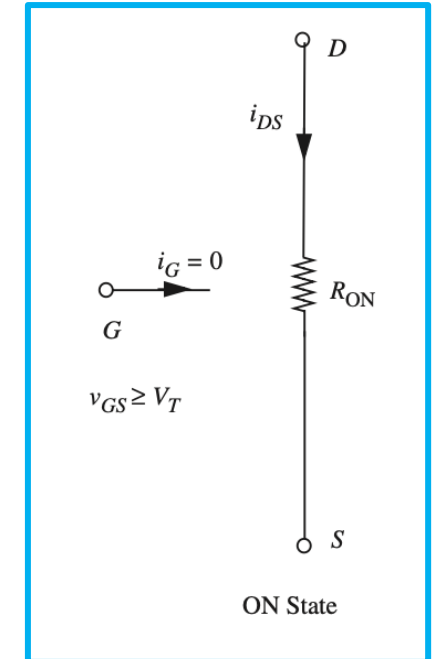
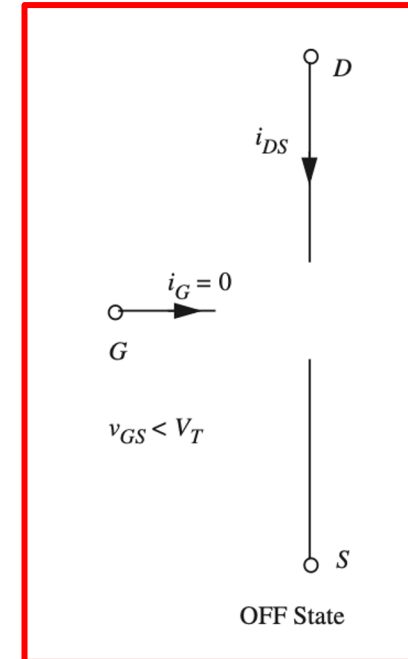
1. **S Mode:** Assumes an ideal channel with zero resistance
2. **SR Model:** Assumes finite channel resistance, R_{ON} , depends on $V_{GS} - V_T = V_{OV}$ ↗ V over drive

MOSFET Linear Models

S Model



SR Model

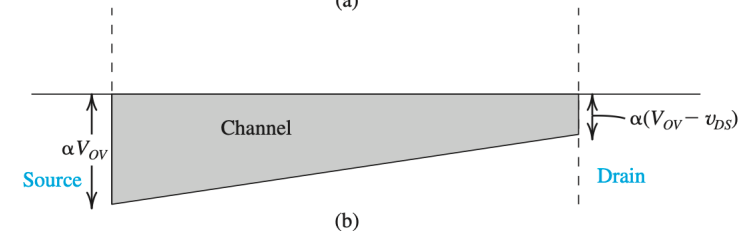
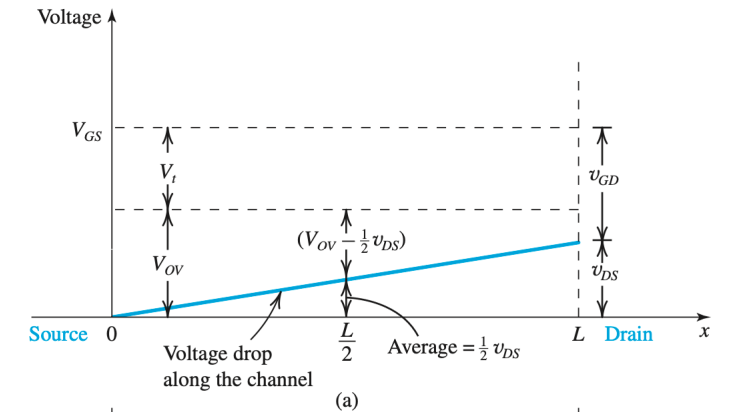
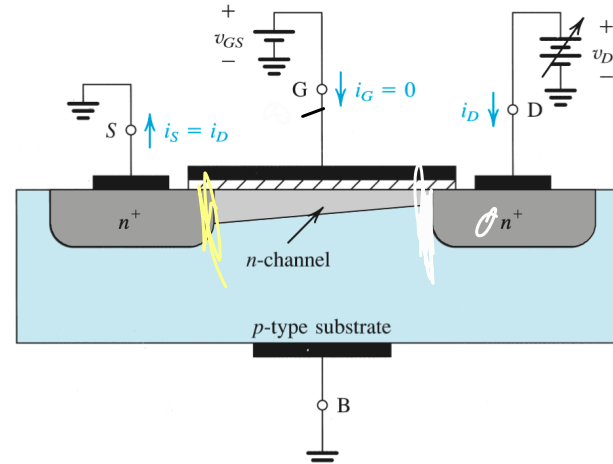
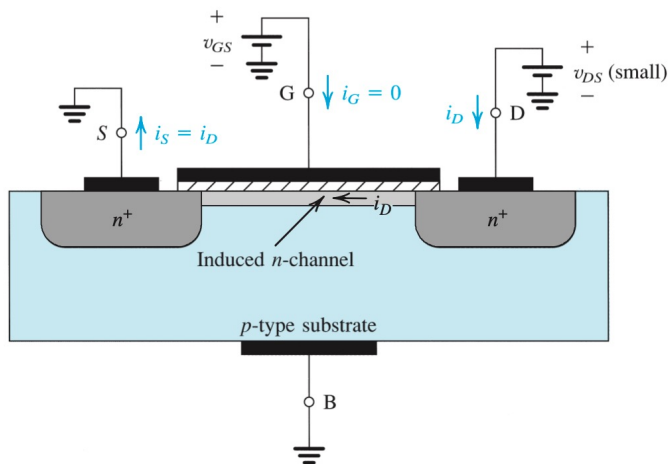


$$R_{ON} = \frac{1}{k'_n \frac{W}{L} (V_{GS} - V_T)} = \frac{1}{k V_{OV}}$$

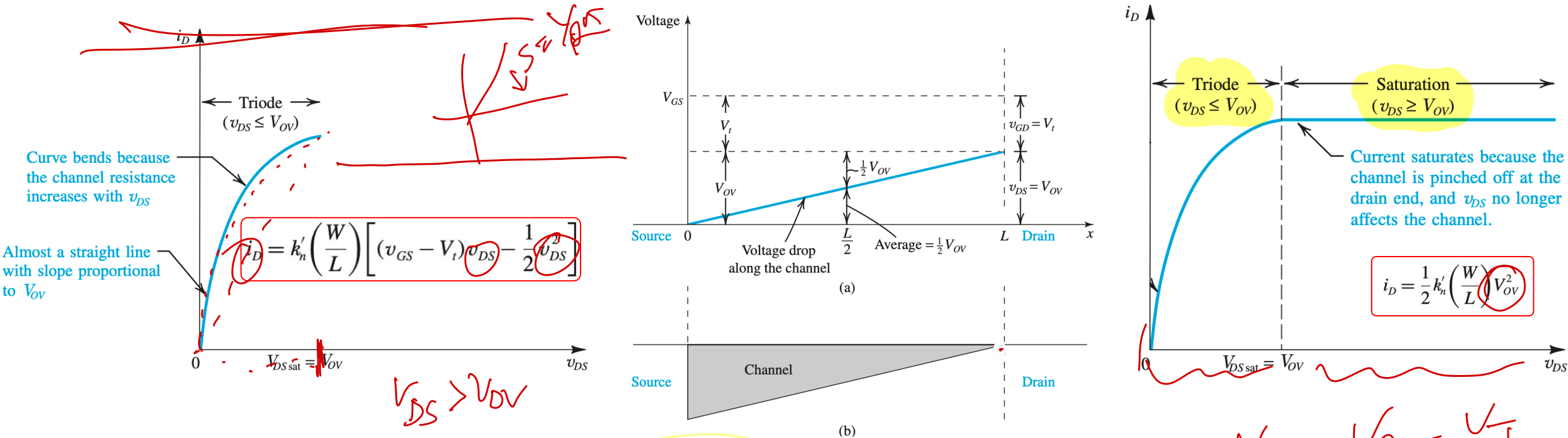
Q1

Real MOSFET

- Why $R_{ON} = \frac{1}{k'_n \frac{W}{L} (V_{GS} - V_T)} = \frac{1}{k V_{OV}}$? Because channel width $\propto V_{OV}$, and $R \propto \frac{1}{\text{width}}$
- For small V_{DS} , uniform channel, hence fixed R_{ON} , therefore SR model valid.
- As V_{DS} is increased, channel becomes tapered cause $V_{GD} \downarrow$. Resistance $\uparrow$, slope $\downarrow$.
- This mode is called the **triode mode**. Condition: $V_{DS} < V_{OV}$

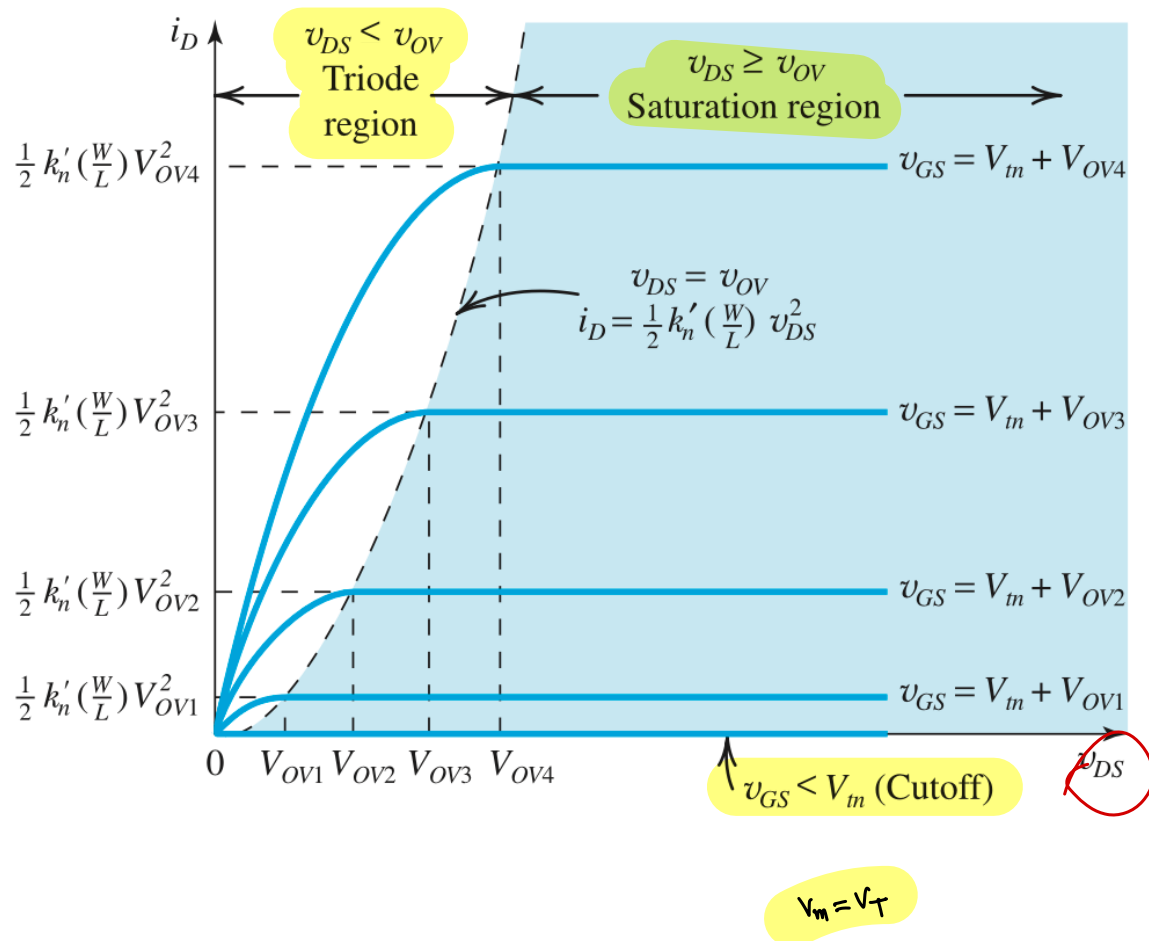


Real MOSFET



- When $V_{DS} = V_{OV}$, channel pinches off.
- Increasing V_{DS} further have no effect on channel shape. Hence, current saturates
- This mode is called the **saturation mode**. Condition: $V_{DS} \geq V_{OV}$
- Behaves like a current source (constant current) that depends on V_{OV}

IV Characteristics of Real MOSFET



Mode	Condition	Equation
Cutoff	$V_{GS} < V_T$	$I_D = 0$
Triode	$V_{GS} \geq V_T$ $V_{DS} < V_{OV}$	$I_D = k[V_{OV}V_{DS} - \frac{1}{2}V_{DS}^2]$
Saturation	$V_{GS} \geq V_T$ $V_{DS} \geq V_{OV}$	$I_D = \frac{k}{2}V_{OV}^2$

$$V_{OV} = V_{GS} - V_T$$

$$k = k'_n \left(\frac{W}{L}\right)$$

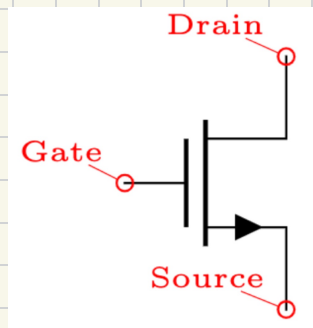
$$I_G = 0$$

Cutoff $\longrightarrow V_{GS} < V_T \quad I_D = 0$

Triode $\longrightarrow \begin{matrix} V_{GS} > V_T \\ V_{DS} < V_{ov} \end{matrix} \quad I_D = k \left[V_{ov} V_{DS} - \frac{1}{2} V_{DS}^2 \right]$

Saturation $\longrightarrow \begin{matrix} V_{GS} > V_T \\ V_{DS} > V_{ov} \end{matrix} \quad I_D = \frac{1}{2} k V_{ov}^2$

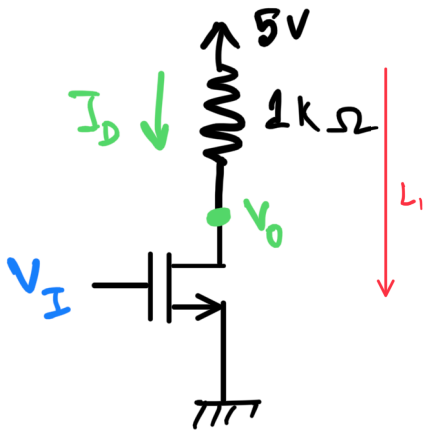
$$V_{ov} = V_{GS} - V_T$$



Solving Circuits with MOSFET

- Use **Method of Assumed State!**
- Three steps:
 - **Assume:** One of the modes (Cutoff, Triode, Saturation)
 - **Solve:** Use corresponding equation and KCL+KVL
 - **Verify:** Check if the conditions of V_{GS} and V_{DS} are satisfied. If not, repeat.
- Might need to solve quadratic equation ($ax^2 + bx + c = 0$).
- If we get two roots, choose the one that's favorable to your assumption

Example 1



The MOSFET is specified as $V_T = 1V$ and $k = 0.5 \text{ mA/V}^2$. Find I_D and V_O for $V_I = 2V$.

Solution:

Step 1: Assume the MOSFET in **saturation**

Step 2: $I_D = \frac{k}{2} V_{OV}^2$ Here, $V_{GS} = V_G - V_S = V_G - 0 = V_G = V_I = 2V$
Therefore, $V_{OV} = V_{GS} - V_T = 2 - 1 = 1V$

$$\therefore I_D = \frac{0.5}{2} (1)^2 = 0.25 \text{ mA}$$

Again, $V_{DS} = V_D - V_S = V_D - 0 = V_D = V_O$

KVL along L_1 : $I_D \times 1k\Omega + V_O = 5 - 0 \Rightarrow V_O = 5 - I_D \times 1k\Omega$

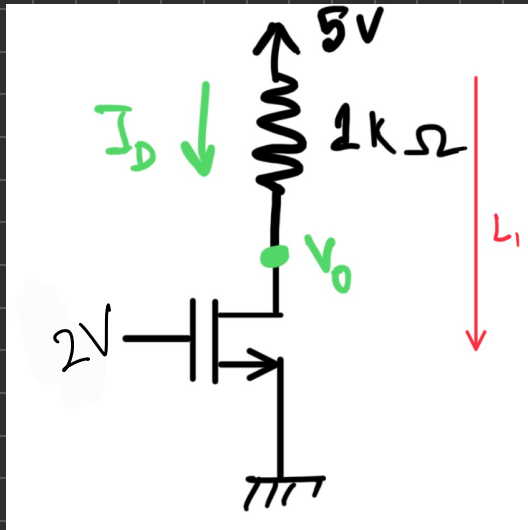
$$\Rightarrow V_O = 5 - 0.25 \times 1 = 4.75 \text{ V} = V_{DS}$$

Step 3: $V_{GS} = 2V > V_T$ ✓ Therefore, **assumption correct!**

$V_{DS} = 1V > V_{OV}$ ✓

Correct ans: $I_D = 0.25 \text{ mA}$, $V_O = 4.75 \text{ V}$

The MOSFET is specified as $V_T = 1V$ and $k = 0.5 \text{ mA/V}^2$. Find I_D and V_O for $V_I = 2V$.



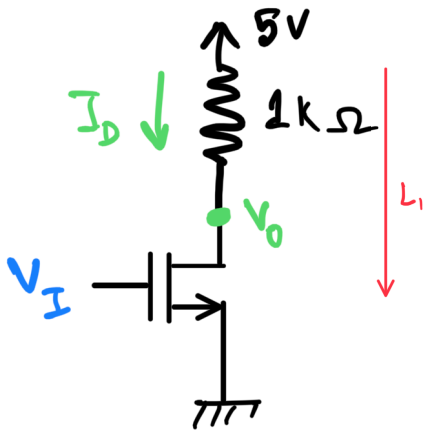
$$\begin{aligned}
 V_{GS} &= V_G - V_S \\
 &= 2 - 0 \\
 \Rightarrow V_{GS} &= 2 \\
 \Rightarrow V_T &= 1V \\
 \Rightarrow V_{OV} &= 2 - 1 = 1V \\
 k &= 0.5 \text{ mA/V}^2
 \end{aligned}$$

$$\begin{aligned}
 V_{DS} &> V_T \\
 \therefore \text{Either Triode or Saturation} \\
 \text{Let's assume,} \\
 \text{Saturation mode} \\
 \therefore I_D &= \frac{1}{2} k V_{OV}^2 \\
 &= \frac{1}{2} \times 0.5 \times 1 \\
 I_D &= 0.25V
 \end{aligned}$$

$$\begin{aligned}
 \frac{5 - V_O}{1} &= I_D \\
 \Rightarrow V_O &= 5 - 0.25 \\
 &= 4.75
 \end{aligned}$$

$$\begin{aligned}
 V_{GS} &> V_T \\
 V_{DS} &> V_{OV} \\
 \therefore 4.75 &> 1 \\
 \therefore \text{Assumption} \\
 &\text{verified.} \\
 &(\text{Ans})
 \end{aligned}$$

Example 2



The MOSFET is specified as $V_T = 1V$ and $k = 0.5 \text{ mA/V}^2$. Find I_D and V_O for $V_I = 5V$.

Solution:

Step 1: Assume the MOSFET in **saturation**

Step 2: $I_D = \frac{k}{2} V_{OV}^2$ Here, $V_{GS} = V_G - V_S = V_G - 0 = V_G = V_I = 5V$
Therefore, $V_{OV} = V_{GS} - V_T = 5 - 1 = 4V$

$$\therefore I_D = \frac{0.5}{2} (4)^2 = 4 \text{ mA}$$

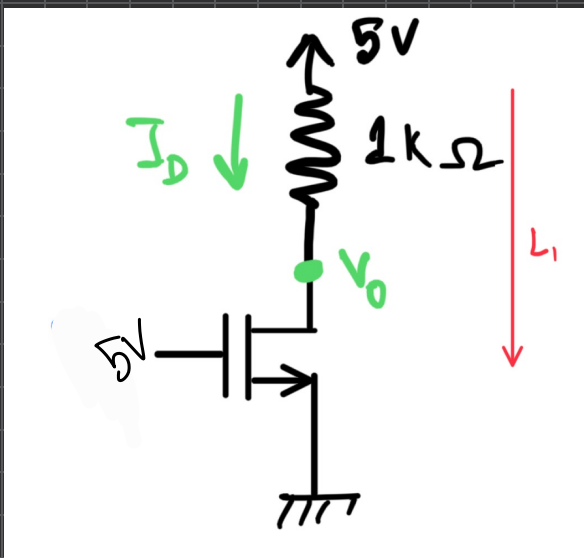
Again, $V_{DS} = V_D - V_S = V_D - 0 = V_D = V_O$

KVL along L_1 : $I_D \times 1k\Omega + V_O = 5 - 0 \Rightarrow V_O = 5 - I_D \times 1k\Omega$

$$\Rightarrow V_O = 5 - 4 \times 1 = 1 \text{ V} = V_{DS}$$

Step 3: $V_{GS} = 5V > V_T$ ✓ Therefore, **assumption wrong!**

$$V_{DS} = 1V \not> V_{OV} \times$$



$$V_{GS} = 5V$$

$$V_T = 1V$$

$$\therefore V_{OV} = 5 - 1 = 4V$$

$$K = 0.5$$

$V_{GS} > V_T \therefore$ Saturation on Triode

Let's, assume saturation.

$$\begin{aligned} \therefore I_D &= \frac{1}{2} K V_{OV}^2 \\ &= \frac{1}{2} \times 0.5 \times 4^2 \\ &= 4 \text{ mA} \end{aligned}$$

$$\therefore 5 - V_D = 4 \times 1$$

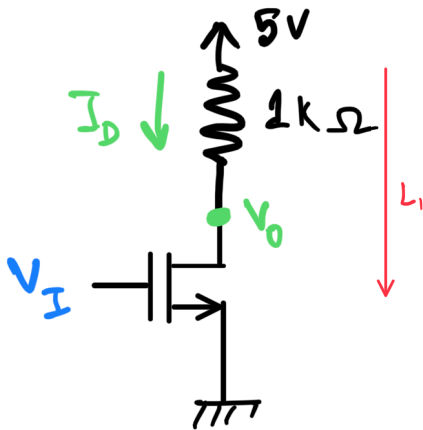
$$\Rightarrow V_D = 1$$

$$\therefore V_{DS} = 1$$

$$\therefore V_{DS} \neq V_{OV}$$

Assumption wrong.

Example 2



The MOSFET is specified as $V_T = 1V$ and $k = 0.5 \text{ mA/V}^2$. Find I_D and V_O for $V_I = 5V$.

Repeat:

Step 1: Assume the MOSFET in **triode**

Step 2: $I_D = k[V_{OV}V_{DS} - \frac{1}{2}V_{DS}^2]$

Here, $V_{GS} = V_G - V_S = V_G - 0 = V_G = V_I = 5V$

Therefore, $V_{OV} = V_{GS} - V_T = 5 - 1 = 4V$

Again, $V_{DS} = V_D - V_S = V_D - 0 = V_D = V_O$. Assuming $V_{DS} = x$

KVL along L_1 : $I_D \times 1k\Omega + V_O = 5 - 0 \Rightarrow I_D = \frac{5 - V_{DS}}{1} = 5 - x$

$$\therefore I_D = 0.5 \left[4 \times V_{DS} - \frac{1}{2} V_{DS}^2 \right] \Rightarrow (5 - x) = 0.5 \left[4x - \frac{1}{2} x^2 \right]$$

$$\Rightarrow 5 - x = 2x - 0.25x^2 \Rightarrow 0.25x^2 - 3x + 5 = 0$$

Solving, $x = 2V, \cancel{x = 10V}$

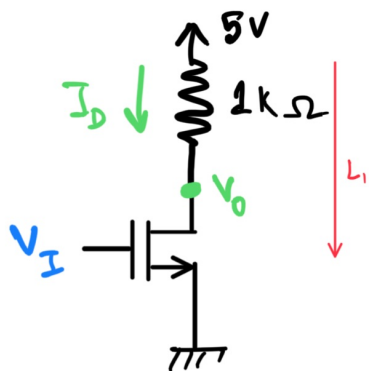
Since $V_{DS} = x$ is small in triode, smaller value of x is favorable

Therefore, $V_O = V_{DS} = x = 2V$, and $I_D = 5 - x = 3 \text{ mA}$

Step 3: $V_{GS} = 5V > V_T$ ✓ Therefore, **assumption correct!**

$V_{DS} = 2V < V_{OV}$ ✓

Correct ans: $I_D = 3 \text{ mA}, V_O = 2 \text{ V}$



The MOSFET is specified as $V_T = 1V$ and $k = 0.5 \text{ mA/V}^2$. Find I_D and V_O for $V_I = 5V$.

Let's assume, Triode,

$$\therefore I_D = k \left[V_{ov} V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

$V_{DS} = V_D - V_S$
 $= V_D - 0$
 $\Rightarrow V_D = x$

$$= 0.5 \left[4x - \frac{1}{2} x^2 \right]$$

$$= -0.25x^2 + 2x$$

Again,

$$\frac{5 - V_D}{1} = I_D = -0.25x^2 + 2x$$

$$\Rightarrow 5 - x = -0.25x^2 + 2x$$

$$\Rightarrow 0.25x^2 - 3x + 5 = 0$$

$$x = 10$$

$$x = 2 \quad \left[\text{we will take this value} \right]$$

$$\therefore V_D = 2$$

$$\therefore V_{DS} = 2, V_{GS} = 5 > V_T, V_{ov} = V_{GS} - V_T$$

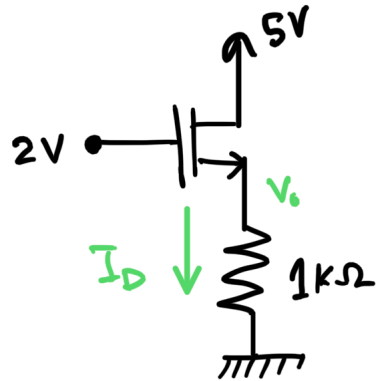
$$= 5 - 1$$

$$= 4$$

$$\therefore V_{DS} < V_{ov}$$

$\therefore$ Assumption correct. Triode mode.

Example 3



The MOSFET is specified as $V_T = 1V$ and $k = 4 \text{ mA/V}^2$. Find I_D and V_o

Solution:

Step 1: Assume the MOSFET in **saturation**

Step 2: $I_D = \frac{k}{2} V_{ov}^2$

Let's assume $V_o = V_S = x$

Here, $V_{GS} = V_G - V_S = V_G - V_o = 2 - x$

Therefore, $V_{ov} = V_{GS} - V_T = (2 - x) - 1 = 1 - x$

Again, $V_{DS} = V_D - V_S = V_D - V_o = 5 - x$

Ohm's law for the resistor: $I_D = \frac{V_o - 0}{1k\Omega} = x$

$$\therefore x = \frac{4}{2} (1 - x)^2 \Rightarrow x = 2(1 - 2x + x^2) \Rightarrow x = 2 - 4x + 2x^2$$

$$\Rightarrow 2x^2 - 5x + 2 = 0$$

Solving, $x = 0.5, \cancel{x = 2V}$

Since $V_{DS} = 5 - x$ is large in saturation smaller value of x is favorable

$\therefore V_o = V_S = x = 0.5V, I_D = x = 0.5 \text{ mA},$
 $V_{DS} = 5 - x = 4.5V, V_{GS} = 2 - x = 1.5V, \text{ and } V_{ov} = 1 - x = 0.5V$

Step 3: $V_{GS} = 1.5V > V_T$ ✓ Therefore, **assumption correct!**

$V_{DS} = 4.5V > V_{ov}$ ✓ **Correct ans: $I_D = 0.5 \text{ mA}, V_o = 0.5 V$**

Practice

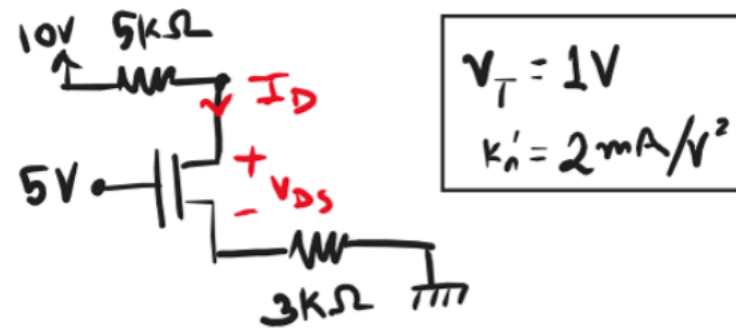
Question 4 [CO1, CO4]

10

Analyze the following circuit to find the values of I_D and V_{DS} **using** the Method of Assumed State. You must **validate** your assumptions.

[7 + 3]

Hint: Use I_D as unknown x . Use Ohm's law to represent V_D and V_S in terms of x .



Hint Explanation

Assume $I_D = x$. For 5kΩ: $I_D = \frac{10 - V_D}{5} \Rightarrow V_D = 10 - 5 \times I_D = 10 - 5x$.

For 3kΩ: $I_D = \frac{V_S - 0}{3} \Rightarrow V_S = 3 \times I_D = 3x$.

Therefore, $V_{GS} = V_G - V_S = 5 - 3x$, and $V_{OV} = V_{GS} - V_T = (5 - 3x) - 1$

Also, $V_{DS} = V_D - V_S = (10 - 5x) - 3x = 10 - 8x$

Now if you assume saturation:

$$I_D = \frac{k}{2} V_{OV}^2 \Rightarrow x = \frac{2}{2} (4 - 3x)^2$$

And if you assume triode:

$$I_D = k [V_{OV} V_{DS} - \frac{1}{2} V_{DS}^2]$$

$$\Rightarrow x = 2[(4 - 3x)(10 - 8x) - 0.5 \times (10 - 8x)^2]$$

Solve for x , take the _____ root

$$x = \frac{2}{2} (4 - 3x)^2$$

$$\Rightarrow x = 16 - 2 \cdot 3x \cdot 4 + 9x^2$$

$$\Rightarrow 9x^2 - 24x - x + 16 = 0$$

$$\Rightarrow 9x^2 - 25x + 16 = 0$$

$$\therefore x = \textcircled{1} \parallel 1.78$$

$\therefore$

$$V_{GS} = 5 - 3x = 2 \quad \left. \begin{array}{l} V_{GS} \geq V_T \\ 2 > 1 \end{array} \right\}$$

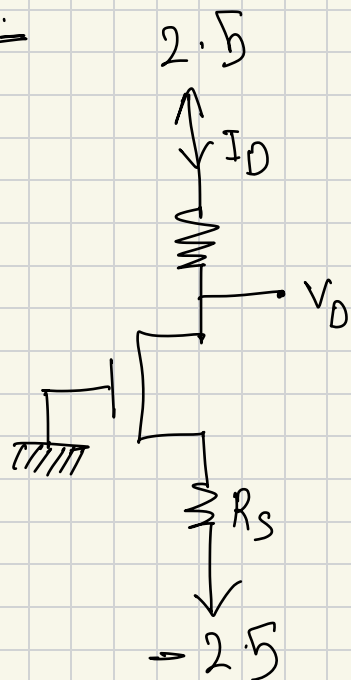
$$V_{OV} = 4 - 3x = 1$$

$$\uparrow V_{DS} = 10 - 8x \downarrow = 2$$

$$V_{DS} > V_{OV} \\ 2 > 1$$

$\therefore$ Assumption correct.

Example:



$$V_T = 0.7V$$

$$K = 3.2 \text{ mA/V}^2$$

$$I_D = 0.4 \text{ mA}$$

$$V_D = 0.5V$$

$$V_S = ?$$

$$R_S = ?$$

$$R_D = \frac{2.5 - 0.5}{0.4} = 5k\Omega$$

$$\therefore V_{GS} = 0 - V_S = -V_S = -x$$

$$V_{OV} = V_{GS} - V_T = -V_S - 0.7 = -x - 0.7$$

$$V_{DS} = 0.5 - V_S = 0.5 - x = 0.5 - x$$

Saturation

$$I_D = \frac{K}{2} V_{ov}^2$$

$$\Rightarrow 0.4 = \frac{3.2}{2} (-x - 0.7)^2$$

$$\therefore x = ?$$

if $x = -0.2$

$$\therefore V_{GS} = 0.2$$

$$V_{ov} = 0.2 - 0.7 = -0.5$$

$$V_{DS} = 0.5 - 0.2 = 0.3$$

$$0.25 = (x + 0.7)^2$$

$$\Rightarrow \pm 0.5 = x + 0.7$$

$$\Rightarrow x = -0.2, -1.2$$

$$V_{DS} > V_{ov}$$

$$0.7 > -0.5$$

$$V_{GS} > V_T$$

$$0.2 \not> 0.7$$

if

$$x = -1.2$$

$$V_{GS} = 1.2$$

$$V_{OV} = 1.2 - 0.7 = 0.5$$

$$V_{DS} = 0.5 + 1.2 = 1.7$$

$$\therefore V_{GS} > V_T$$

$$1.2 > 0.7$$

$$V_{DS} > V_{OV}$$

$$1.7 > 0.5$$

$$\therefore V_S = -1.2$$

$$\therefore R_S = \frac{-1.2 - (-2.5)}{0.4} = 3.25 \text{ k}\Omega \quad (\text{Ans})$$

Diode

1st থাকলে 2nd state

2nd " 4th "

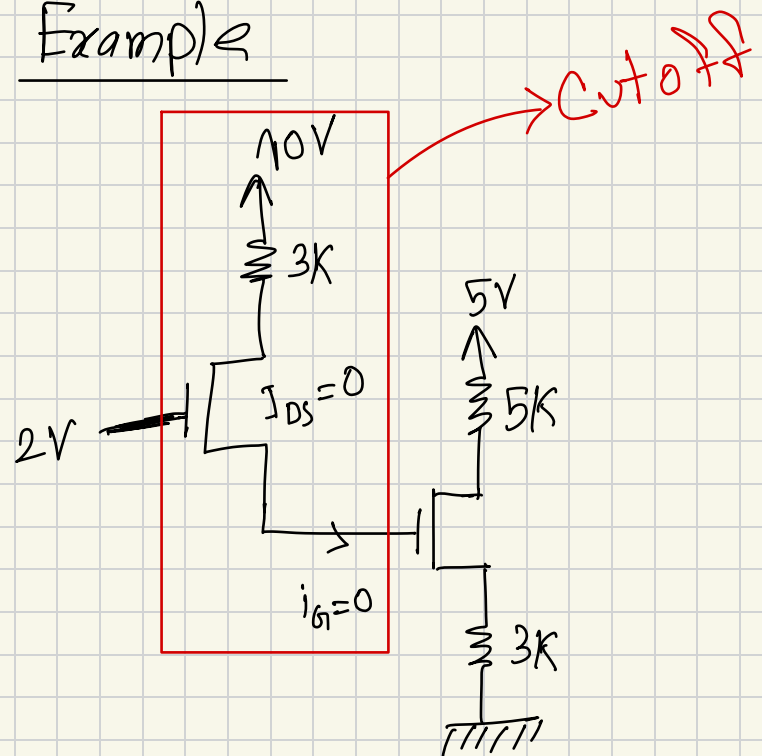
Mosfet

1 → 3

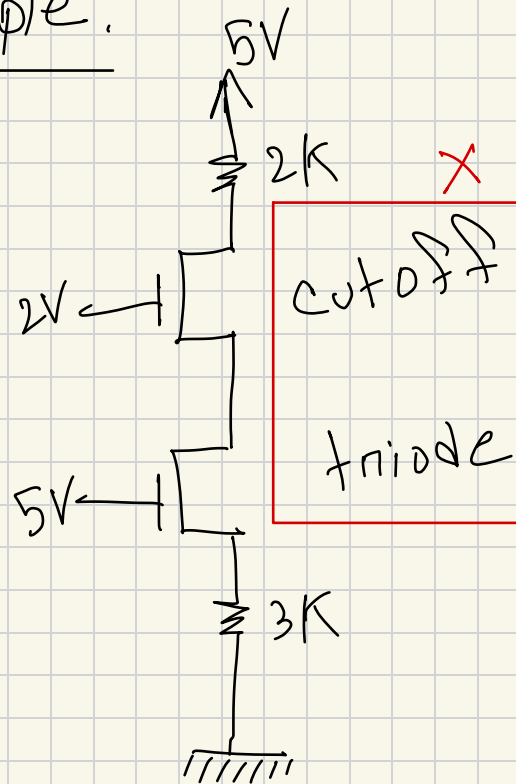
2 → 9

3 → 27

Example



Example:



cutoff

triode

cutoff

cutoff

triode

triode/saturation

saturation

triode/saturation

Assignment:

Deadline: 19 Sept

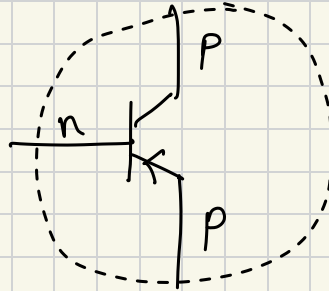
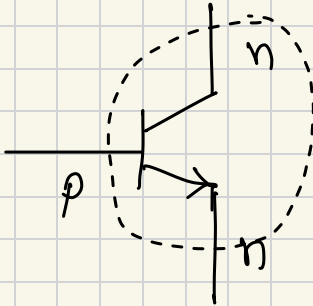
Quiz: 03

Sunday, 15 Sept

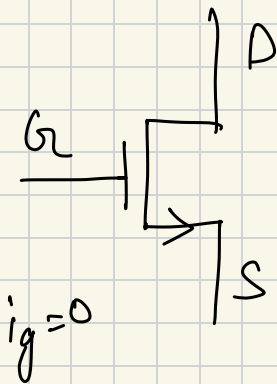
MOSFET

BJT

(Bipolar Junction Transistor)

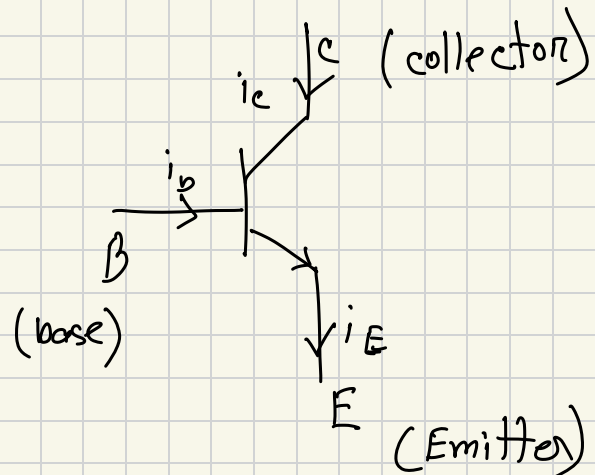


MOSFET



$$I_D = I_S = I_{DS}$$

BJT



$$i_B \neq 0 \text{ (In general)}$$

অফ শালীন $i_B \approx 0$

$$\therefore i_B \neq 0$$

$$i_E = i_B + i_C$$

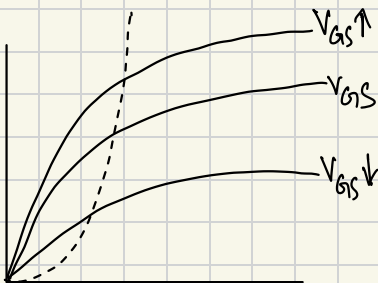
$$\therefore i_C \neq i_E$$

(have to use this formula)

Control Terminal (G)

Voltage controlled (V_{GS})

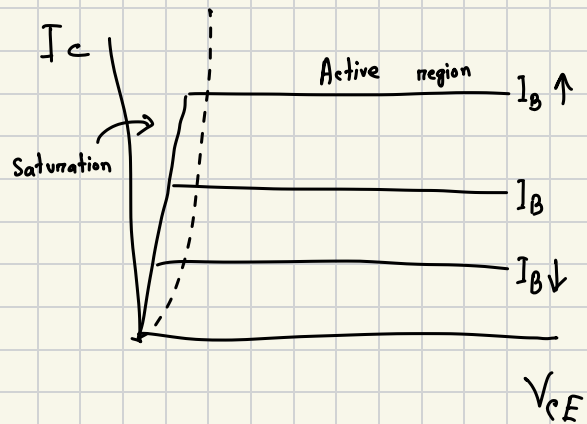
V_{DS} vs I_{DS}



Control Terminal (B)

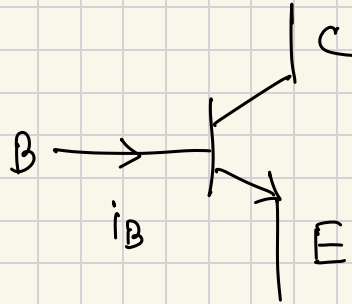
Current controlled (I_B)

I_C vs V_{CE}



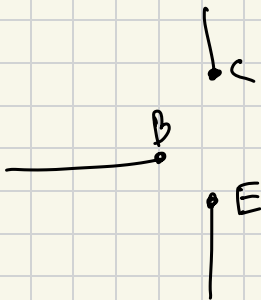
BJT S-model

same as MOSFET



BJT OFF

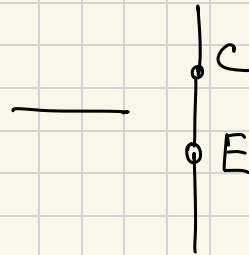
$$i_B = 0 \text{ (open circuit)}$$



$$\therefore i_C = 0$$

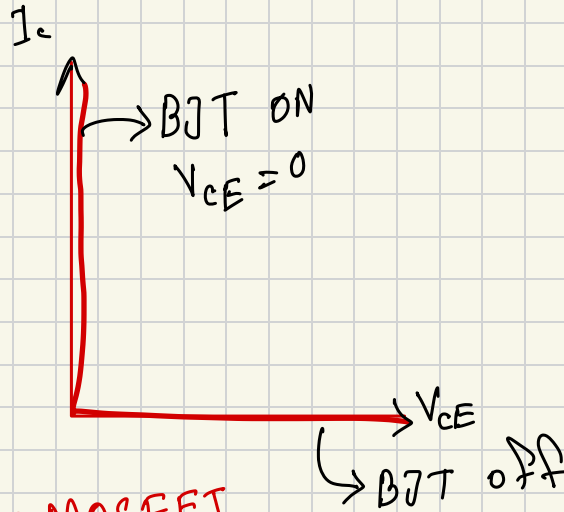
BJT ON

$$i_B \geq i_{th} (0.2 \text{ mA})$$



short circuit

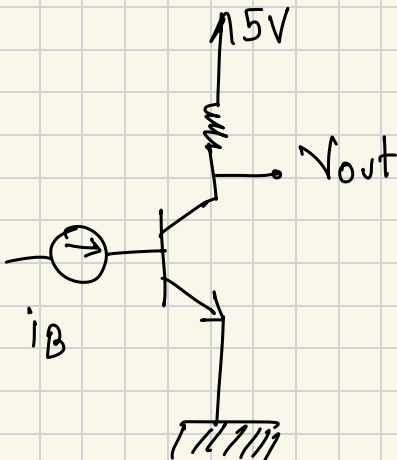
$$V_{CE} = 0$$



Same as MOSFET

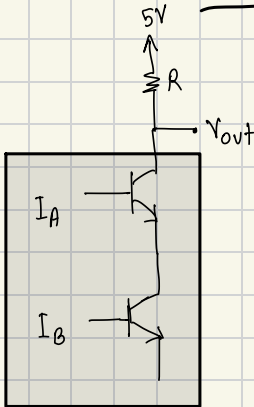
Logic Gate with BJT

NOT



	V_{out}
$I_B = 0$ (0)	5V 1
$I_B \geq I_{th}$ (1)	0V (0)

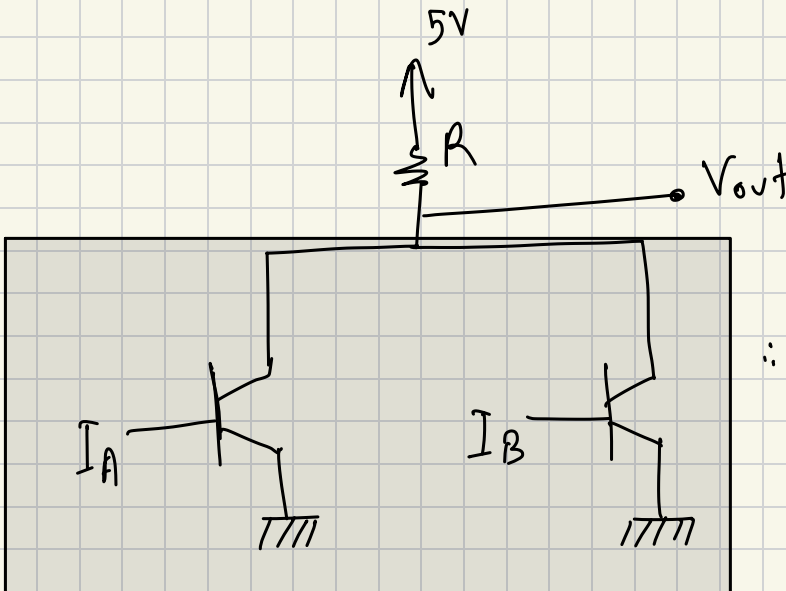
NAND



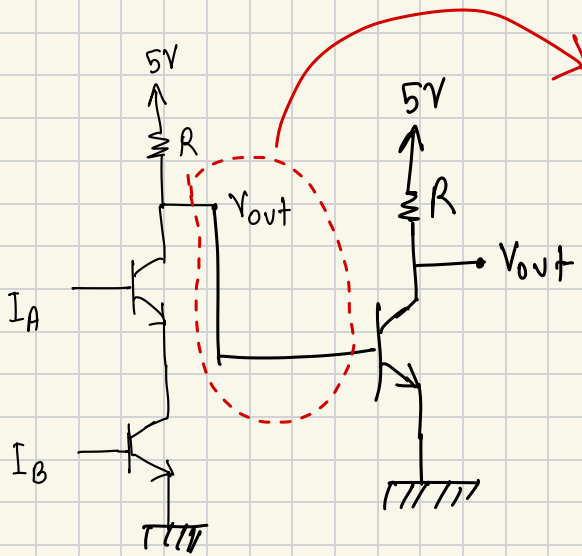
$$V_{out} = \overline{I_A I_B}$$

$$\overline{A+B} = \bar{A} \cdot \bar{B}$$
$$\therefore \overline{AB} = \bar{A} + \bar{B}$$

NOR



$$\therefore V_{out} = \overline{I_A + I_B}$$

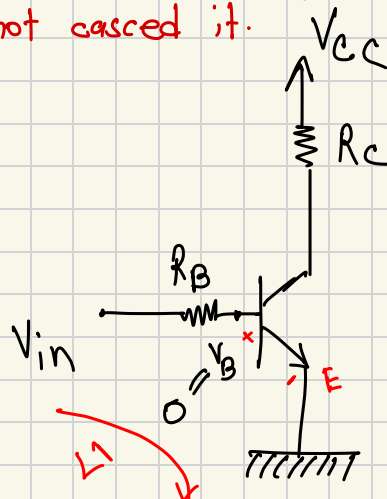
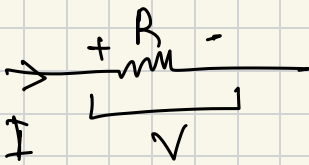


Input takes Current,
but input voltage is V .

So, we can not
cascad BJT for
logical operation

controlling BJT with Voltage

We are trying to control a BJT because
it takes input as current and gives output as
voltage. thats why we can not cascad it.



Normally, (We know this from BJT's Switch Model)

$$I_B \approx 0 \quad \text{or} \quad V_{out} = 1$$

$$I_B \geq I_{th} \quad \text{or} \quad V_{out} = 0$$

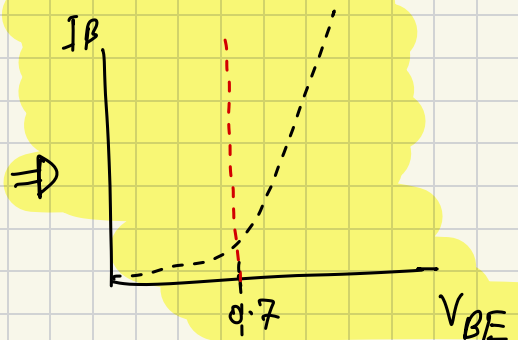
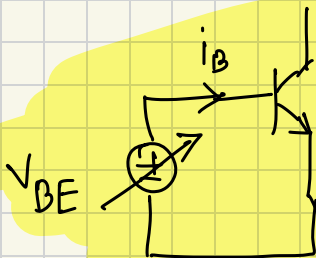
For L1,

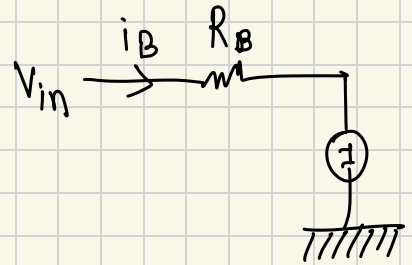
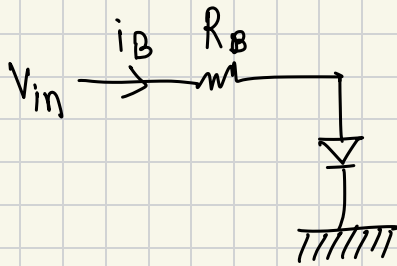
$$V_{in} - 0 = i_B R_B + V_{BE}$$

$$\Rightarrow i_B = \frac{V_{in} - V_{BE}}{R_B}$$

$$V_{BE} = f(I_B)$$

2 way lock.
It's a loop.





Diode OFF

$$V_{in} < 0.7$$

$$\Rightarrow i_B = 0$$

Diode ON

$$V_{in} - 0 = i_B R_B + 0.7$$

$$V_{in} - 0.7 = i_B R_B$$

$$\Rightarrow V_{in} = 0.7 + i_B R_B$$

$$\Rightarrow V_{th} = i_{th} R_B + 0.7$$

$$V_{in} < 0.7 \leftarrow i_B = 0$$

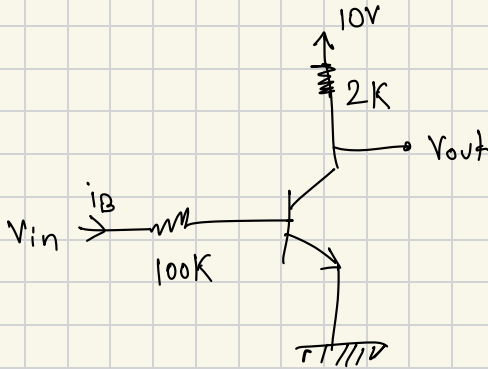
$$V_{out} = 1$$

$$V_{in} \geq V_{Th} \leftarrow i_B \geq i_{th}$$

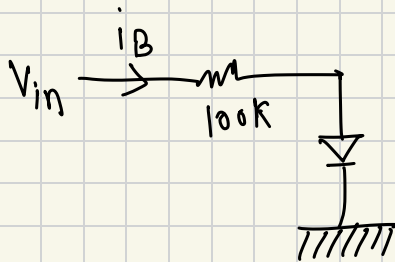
$$V_{out} = 0$$

$$f(R_B, i_{th})$$

Ex



(?) find the value of V_{in} for which $i_B \geq i_{th}$, $i_{th} = 0.2mA$



<u>OFF</u> $V_{in} < 0.7$ $\Rightarrow i_B = 0$	<u>ON</u> $V_{in} - 0.7 = i_B \times 100K$
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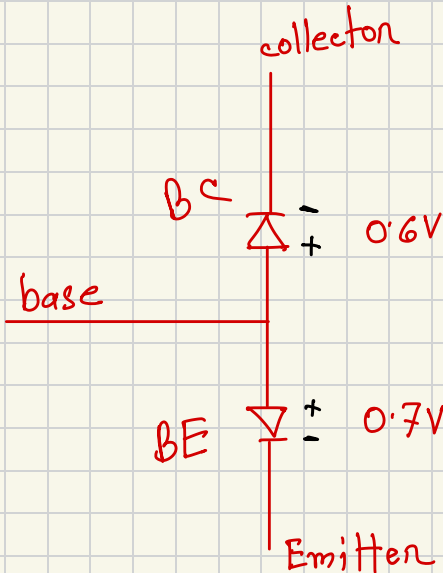
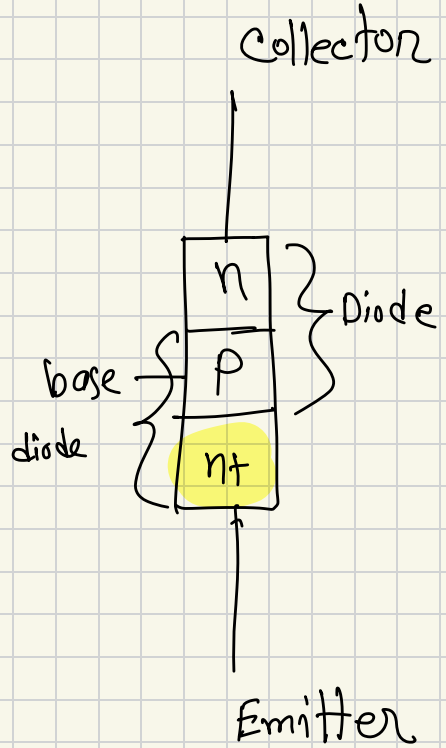
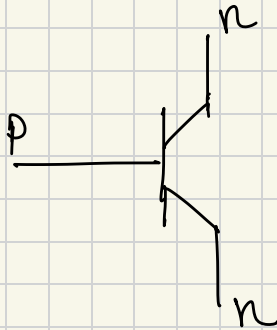
$$\Rightarrow V_{th} = 0.7 + i_{th} \times 100K$$

$$V_{th} = 0.2 \times 100 + 0.7$$

$$V_{th} = 20.7V \quad (Ans)$$

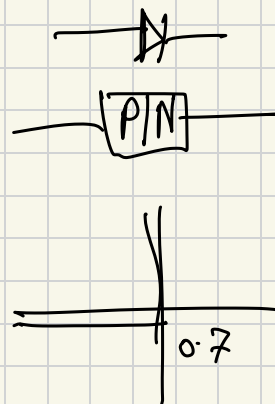
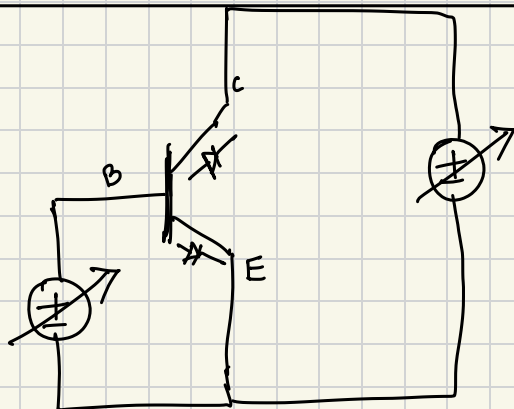
BJT

Diode model



doping different
হওয়ায় কারণে প্রভাব

$$I_E = I_B + I_C$$



BE diode (OFF)

$$V_{BE} < 0.7$$

BC diode (OFF)

$$V_{BC} < 0.6$$

$$\Rightarrow V_B - V_C < 0.6$$

$$\Rightarrow V_B - V_E - (V_C - V_E) < 0.6$$

$$\Rightarrow V_{BE} - V_{CE} < 0.6$$

$$\Rightarrow V_{CE} > V_{BE} - 0.6$$

$$\Rightarrow V_{BE} < V_{CE} + 0.6$$

States of diode :

BE $\rightarrow$ ON, OFF

BC $\rightarrow$ ON, OFF

	BE	BC
cutoff	OFF	OFF
saturation	ON	ON
Active	ON	OFF
Reverse active	OFF	ON

X $\rightarrow$ not for this course

Case I

BE OFF

BC OFF

$$V_{BE} < 0.7$$

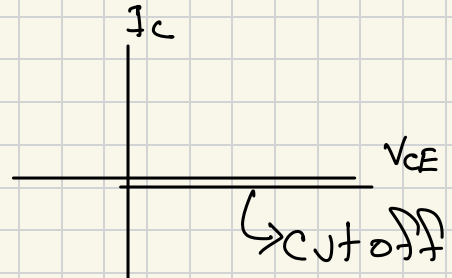
$$V_{CE} > V_{BE} - 0.6$$

cutoff

$$I_C = 0$$

$$I_B = 0$$

$$I_E = 0$$



Case II

Active

BE ON

BC OFF

$$I_B > 0$$

$$V_{CE} > V_{BE} - 0.6$$

$$I_c = \beta I_B$$

$$\Rightarrow \beta = \frac{I_c}{I_B}$$

$\therefore$ BJT acts as an amplifier.

$$I_E = I_B + I_c$$

$$= I_B + \beta I_B$$

$$I_E = I_B(1 + \beta)$$

$$I_E = I_B + I_c$$

$$\Rightarrow I_E = \frac{I_c}{\beta} + I_c$$

$$= \left(1 + \frac{1}{\beta}\right) I_c$$

$$I_E = \left(\frac{\beta + 1}{\beta}\right) I_c$$

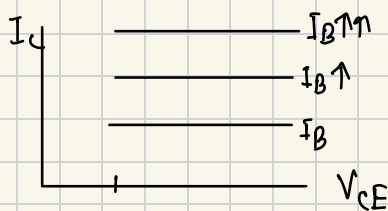
$$\Rightarrow I_c = \left(\frac{\beta}{1 + \beta}\right) I_E$$

$$\Rightarrow I_c = \alpha I_E$$

যদি active mode এ থাকে তাহলে
collection current, base current এর থেকে
অনেক বড় হবে

emitter current, base current প্রায় same
হবে।

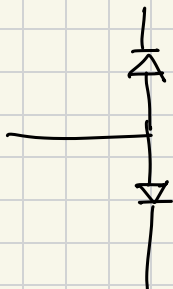
$$I_E = I_B + I_C$$



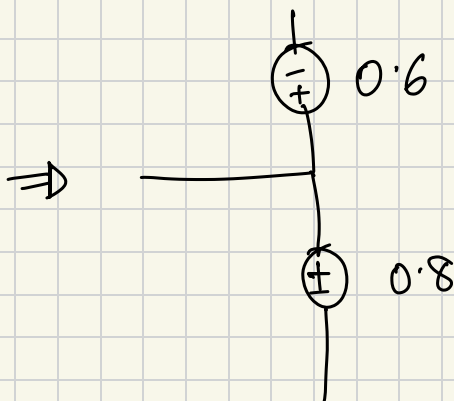
Saturation

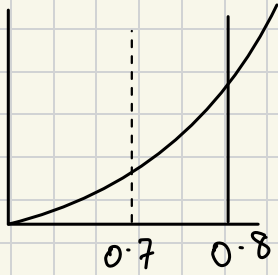
2 টা diode স on.

BE ON

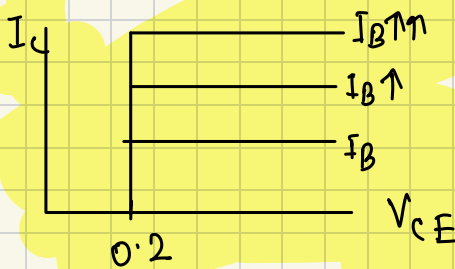


BC ON





current অনেক বেশি flow
হয় B and E দিয়ে।
তাই 0.8 V



In Active,

$$I_C = \beta I_B$$

$$\Rightarrow \beta = \frac{I_C}{I_B}$$

In saturation,

$$I_C \downarrow$$

$$I_B \uparrow$$

$$\frac{I_C}{I_B} < \beta$$

BJT Method of Assumption

① Assumption

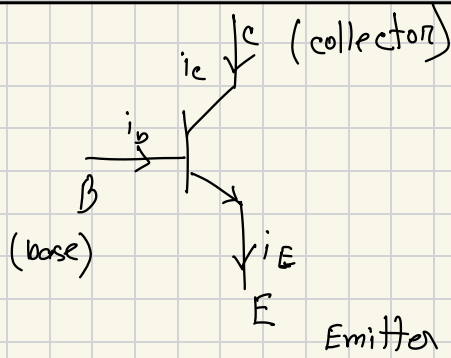
- Active $[V_{BE} = 0.7, I_C = \beta I_B]$
- Saturation $[V_{BE} = 0.8, V_{CE} = 0.2]$
- Cutoff $[I_C = I_B = I_E = 0]$

② Solve $[KCL, KVL, Nodal]$

③ Verification:

- Active $[V_{CE} > 0.2]$
- Saturation $[\frac{I_C}{I_B} < \beta]$
- Cutoff $[V_{BE} < 0.7, V_{BC} < 0.5]$

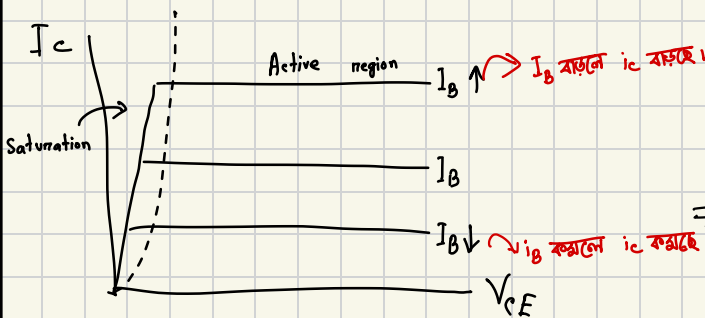
Summary over BJT



BJT দিয়ে আমরা mainly i_b use করে collector and emitter এর মধ্যে current flow control করে।

$i_b \neq 0$, if $i_b = 0$ then the BJT is in cutoff mode.

I_c vs V_{CE}

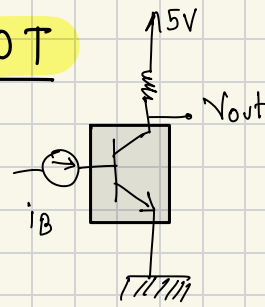


BJT S-Model (Switch Model)

BJT OFF	BJT ON
$i_b = 0$ (open circuit)	$i_b \geq i_{th}$ (0.2mA)
$\therefore i_c = 0$	$V_{CE} = 0$

Logic Gate with BJT

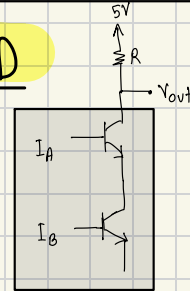
NOT



$I_B = 0$	5V
$I_B \geq I_{th}$	0V

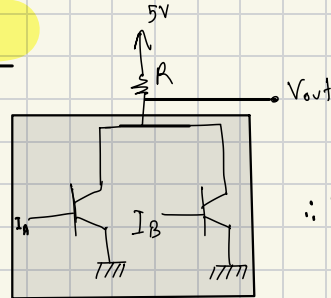
I_B	V_{out}
0	1
1	0

NAND



$$V_{out} = \overline{I_A I_B}$$

NOR



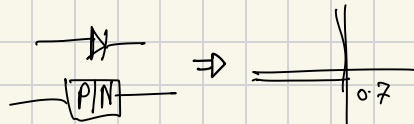
$$\therefore V_{out} = \overline{I_A + I_B}$$

BJT input নেয় current, but output দেয় voltage তাই cascaded করে logical operation করা যায় না।

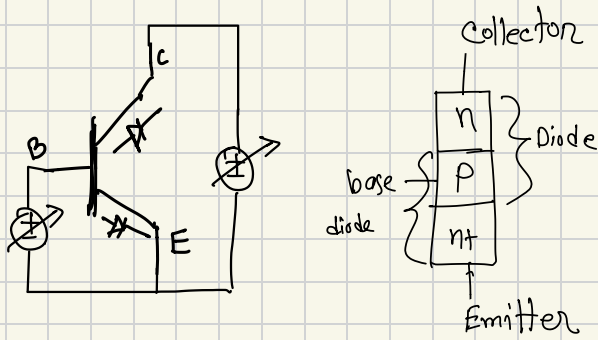
তাই voltage দিয়ে BJT কে control করতে base এর সাথে একটি resistance লাগানো হয়।

#

Diode OFF	Diode ON
$V_{in} < 0.7$ $\therefore i_B = 0$	$V_{in} - 0 = i_B R_B + 0.7$ $\Rightarrow V_{in} = i_B R_B + 0.7$ $\therefore V_{th} = i_{th} R_B + 0.7$
$i_B = 0$ $V_{out} = 1$	$i_B \geq i_{th}$ $V_{out} = 0$



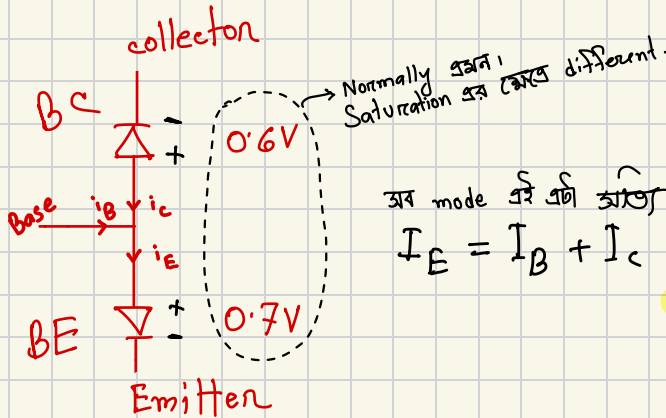
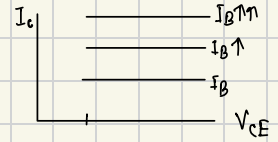
BJT (Diode Model)



In Active mode, BJT is an amplifier.

BE = ON
BC = OFF

① $I_c = \beta I_B$
② $I_c = \alpha I_E$
③ $\alpha = \frac{\beta}{1 + \beta}$
④ $I_E = I_c + I_B$



অব mode এই এটা অর্থাৎ
 $I_E = I_B + I_c$

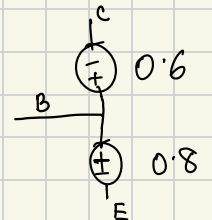
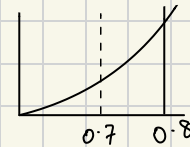
In Active mode,

- * I_c অনেক বড় হয় I_B এর তুলনায়।
- * I_E and I_B প্রায় same থাকে।

In Saturation: 2টা diode-ই ON

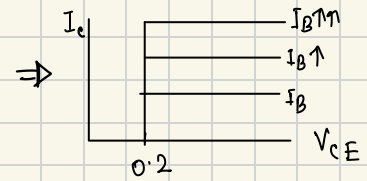
BE = ON
BC = ON

- * B & E এর ভেতর দিয়ে অনেক বেশি current flow হয়। তাই BE এর threshold voltage বেড়ে 0.8 হয়ে যায়।



* So, In Saturation the voltages will be

* In saturation, 0.2V পর্যন্ত $I_c = 0$, তারপর I_B এর কারণে change হয়।



* In saturation,

$$\frac{I_c}{I_B} < \beta$$

BJT Method of Assumption

- ① Assumption
- Cutoff [$I_c = I_B = I_E = 0$]
 - Active [$V_{BE} = 0.7, I_c = \beta I_B$]
 - Saturation [$V_{BE} = 0.8, V_{CE} = 0.2$]

② Solve [KCL, KVL, Node]

③ Verification:

- Cutoff [$V_{BE} < 0.7, V_{BC} < 0.5$]
- Active [$V_{CE} > 0.2$]
- Saturation [$\frac{I_c}{I_B} < \beta$]

অব mode এই এটা অর্থাৎ
 $I_E = I_B + I_c$

	Assumption	Prove
Cutoff	$I_c = I_B = I_E = 0$	$V_{BE} \leq 0.7$ $V_{BC} \leq 0.5$
Active	$V_{BE} = 0.7$ $I_c = \beta I_B$	$V_{CE} > 0.2$
Saturation	$V_{BE} = 0.8$ $V_{CE} = 0.2$	$\beta < \frac{I_c}{I_B}$

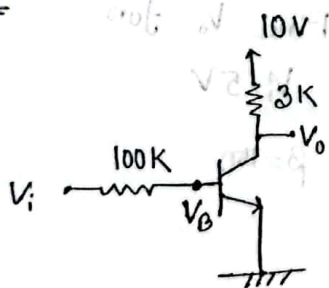
Method of Assumed State:

- (i) Assume — $\begin{cases} \text{Active } [V_{BE} = 0.7V, I_C = \beta I_B] \\ \text{Saturation } [V_{BE} = 0.8V, V_{CE} = 0.2V] \\ \text{Cutoff } [I_B = I_C = I_E = 0] \end{cases}$

(ii) Solve [KCL, KVL, Nodal]

- (iii) Verify — $\begin{cases} \text{Active } [V_{CE} > 0.2V] \\ \text{Saturation } [I_C/I_B < \beta] \\ \text{Cutoff } [V_{BE} \leq 0.7V \text{ and } V_{BC} \leq 0.5V] \end{cases}$

① Example 1:



Find V_o for

$$V_i = 1V$$

$$\beta = 100$$

Assume: Active

$$\textcircled{i} V_{BE} = 0.7V$$

$$\textcircled{ii} I_C = \beta I_B$$

$$I_B = \frac{V_i - V_B}{100K}$$

$$= \frac{1 - 0.7}{100K}$$

$$= \frac{0.3}{100K}$$

$$= 0.003mA$$

$$= 0.003mA$$

$$V_{BE} = 0.7V$$

$$\Rightarrow V_B - V_E = 0.7V$$

$$\Rightarrow V_B - 0 = 0.7$$

$$\Rightarrow V_B = 0.7$$

$$I_C = \beta I_B$$

$$= 100 \times 0.003$$

$$= 0.3mA$$

$$I_c = \frac{10 - V_c}{3}$$

$$\Rightarrow 0.3 = \frac{10 - V_c}{3}$$

$$\Rightarrow V_c = 9.1 \text{ V}$$

$$V_{CE} = V_c - V_E$$

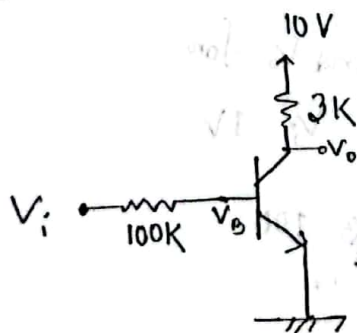
$$= 9.1 - 0$$

$$V_o = 9.1 \text{ V}$$

$$V_{CE} = 9.1 \text{ V} > 0.2$$

Assumption correct.

Example 2:



Find V_o for

$$V_i = 5 \text{ V}$$

$$\beta = 100$$

Assume Active.

$$(i) V_{BE} = 0.7 \text{ V}$$

$$(ii) I_C = \beta I_B$$

$$V_{BE} = 0.7$$

$$I_B = \frac{V_i - V_B}{100}$$

$$I_C = 100 \times I_B$$

$$V_i = 100 \times 0.043$$

$$\Rightarrow V_B - V_E = 0.7$$

$$= \frac{5 - 0.7}{100}$$

$$I_C = 4.3 \text{ mA}$$

$$\Rightarrow V_B = 0.7$$

$$= 0.043 \text{ mA}$$

$$I_c = \frac{10 - V_c}{3}$$

$$\Rightarrow 4.3 = \frac{10 - V_c}{3}$$

$$\Rightarrow V_c = -2.9V$$

$$V_{CE} = V_c - V_E$$

$$= -2.9 - 0$$

$$= -2.9 \neq 0.2$$

Assumption wrong.

Assume saturation.

$$(i) V_{BE} = 0.8V$$

$$(ii) V_{CE} = 0.2V$$

$$V_{BE} = V_B - V_E$$

$$\Rightarrow 0.8 = V_B - 0$$

$$\Rightarrow V_B = 0.8$$

$$I_B = \frac{5 - 0.8}{100}$$

$$= 0.042 \text{ mA}$$

$$V_{CE} = V_c - V_E$$

$$\Rightarrow 0.2 = V_c - 0$$

$$\Rightarrow V_c = 0.2V$$

$$I_c = \frac{10 - 0.2}{3}$$

$$= \frac{9.8}{3}$$

$$= 3.29 \text{ mA}$$

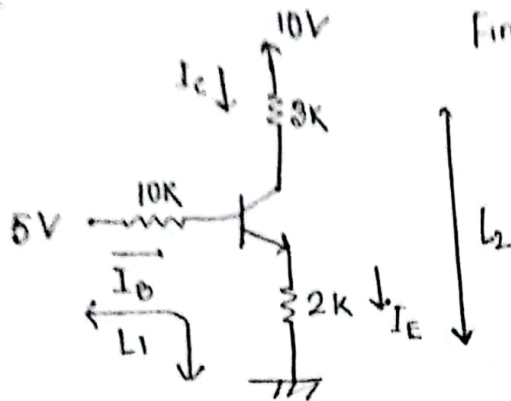
$$\frac{I_c}{I_B} = \frac{3.29}{0.042} = 78.33 < \beta$$

$$V_{CE} = 0.2V$$

Assumption correct.



Example 3:



Find I_B , I_C , I_E , V_{CE} .

① Assume Active.

$$V_{BE} = 0.7V$$

$$I_C = \beta I_B, I_E = \alpha I_E, \alpha = \frac{\beta}{1+\beta}$$

$$1 = \frac{\beta}{\alpha} \frac{I_B}{I_E} \Rightarrow I_E = \frac{\beta}{\alpha} I_B$$

$$5 - 0 = 10I_B + V_{BE} + 2I_E$$

$$\Rightarrow 5 = 10I_B + 0.7 + 2 \times \frac{\beta}{\alpha} I_B$$

$$\Rightarrow 5 = 10I_B + 0.7 + 2 \times \frac{\beta}{1+\beta} I_B$$

$$\Rightarrow 5 = 10I_B + 0.7 + 2 \times (1+\beta) I_B$$

$$\Rightarrow 5 = 10I_B + 0.7 + 2 \times (1+100) I_B$$

$$\therefore I_B = 0.02 \text{ mA}$$

$$I_C = \beta I_B$$

$$= 100 \times 0.02$$

$$= 2 \text{ mA}$$

$$I_E = I_B + I_C$$

$$= 0.02 + 2$$

$$= 2.02 \text{ mA}$$

$$I_E = \frac{V_T - 0}{2}$$

$$\Rightarrow V_T = 2 \times 2.02$$

$$= 4.04V$$

$$I_C = \frac{10 - V_C}{3}$$

$$\Rightarrow 2 = \frac{10 - V_C}{3}$$

$$\Rightarrow V_C = 4V$$

$$V_{CE} = V_C - V_E$$

$$= 4 - 4.04$$

$$= -0.04V \neq 0.2V$$

Wrong assumption.

Assume Saturation.

$$(i) V_{BE} = 0.8V$$

$$(ii) V_{CE} = 0.2V$$

$$I_E = I_C + I_B$$

KVL L1:

$$5 - 0 = 10I_B + 0.8 + 2I_E \quad \text{--- (i)}$$

KVL L2:

$$10 - 0 = 3I_C + 0.2 + 2I_E \quad \text{--- (ii)}$$

$$I_E = I_C + I_B \quad \text{--- (iii)}$$

$$I_B = 0.03 \text{ mA}$$

$$I_C = 1.95 \text{ mA}$$

$$I_E = 1.98 \text{ mA}$$

$$\frac{I_C}{I_B} = \frac{1.95}{0.03}$$

$$= 65 < \beta$$

Assumption correct.